

BFS Traversal

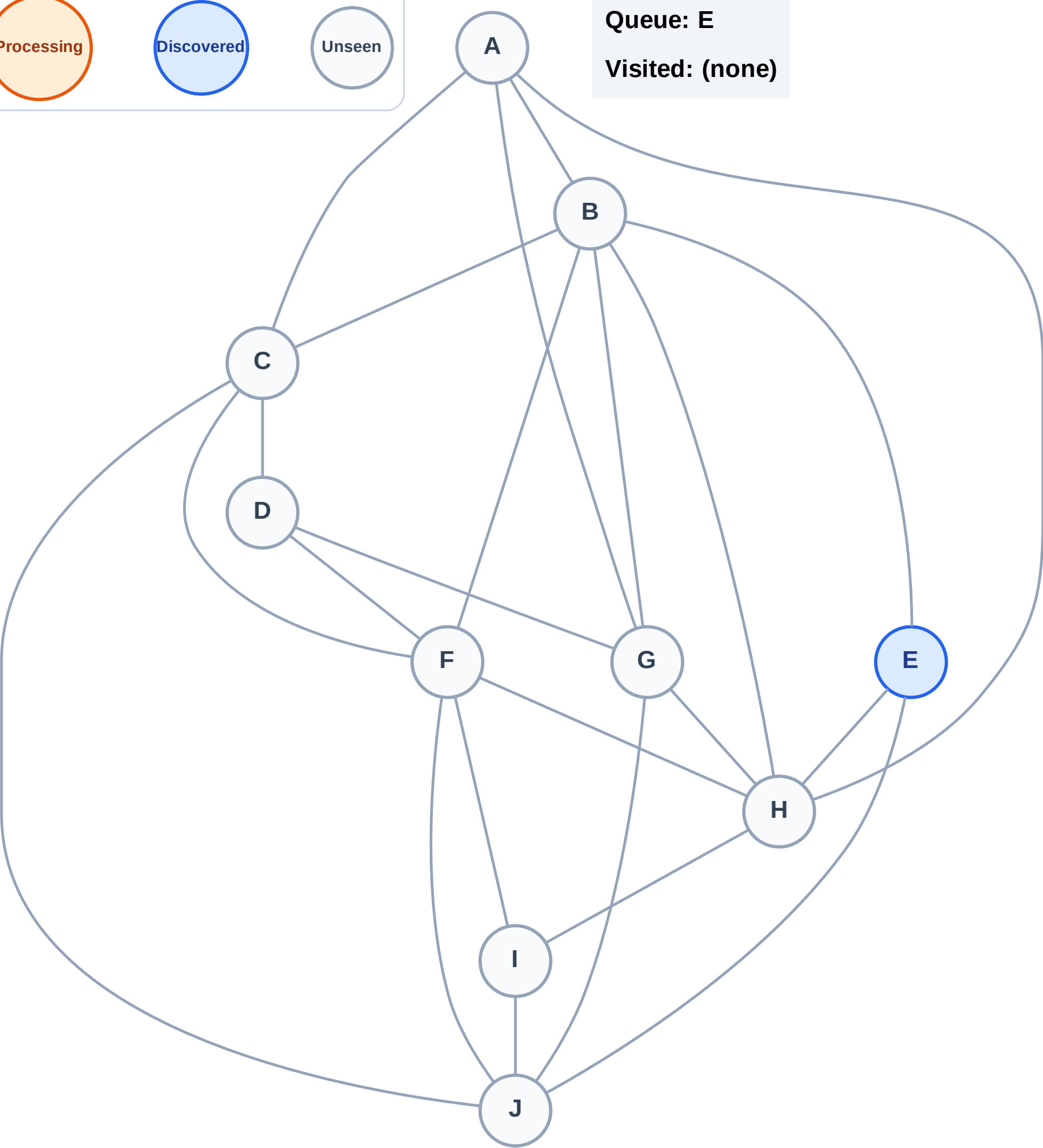
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

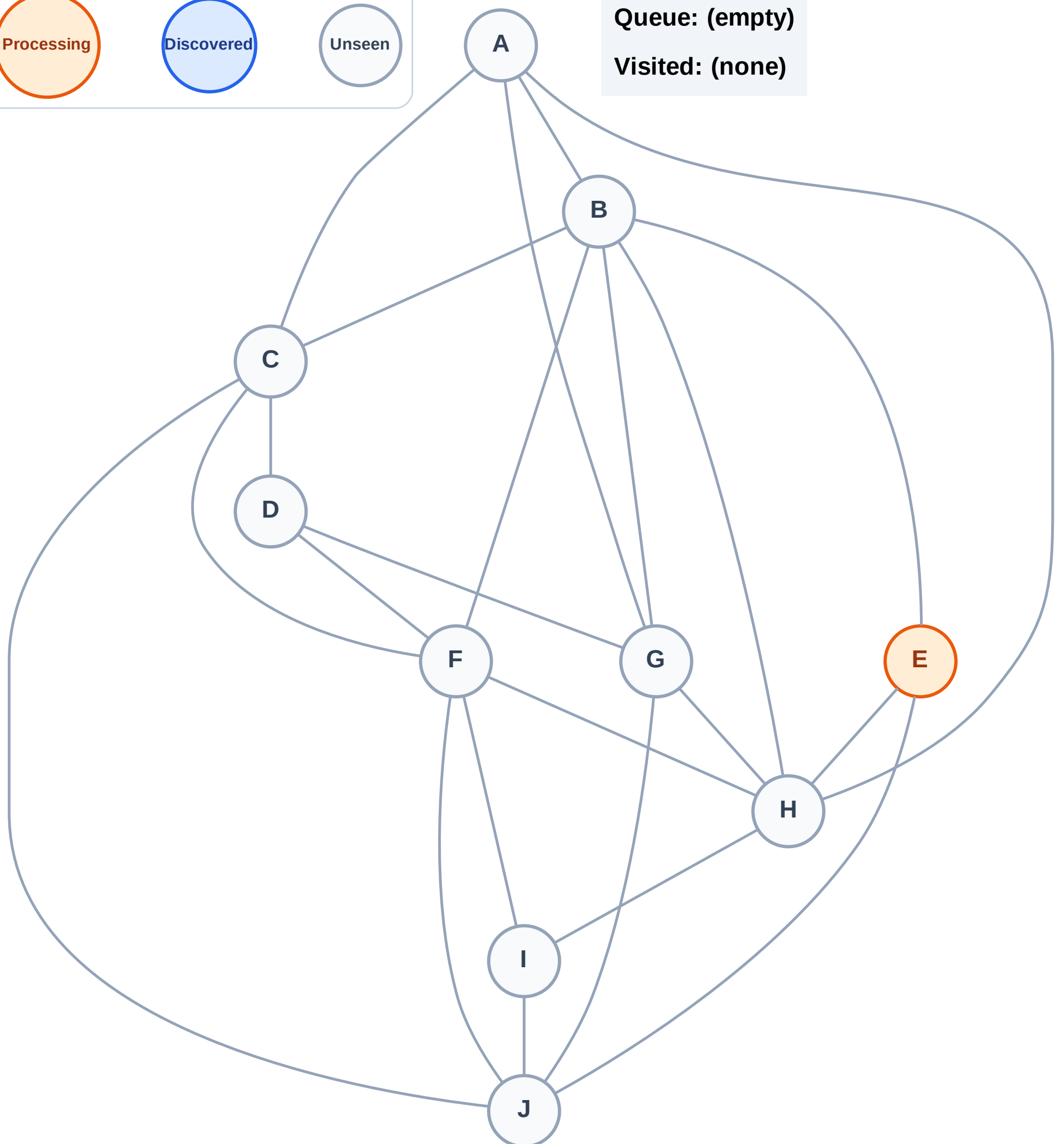
Step 2: Process node E

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

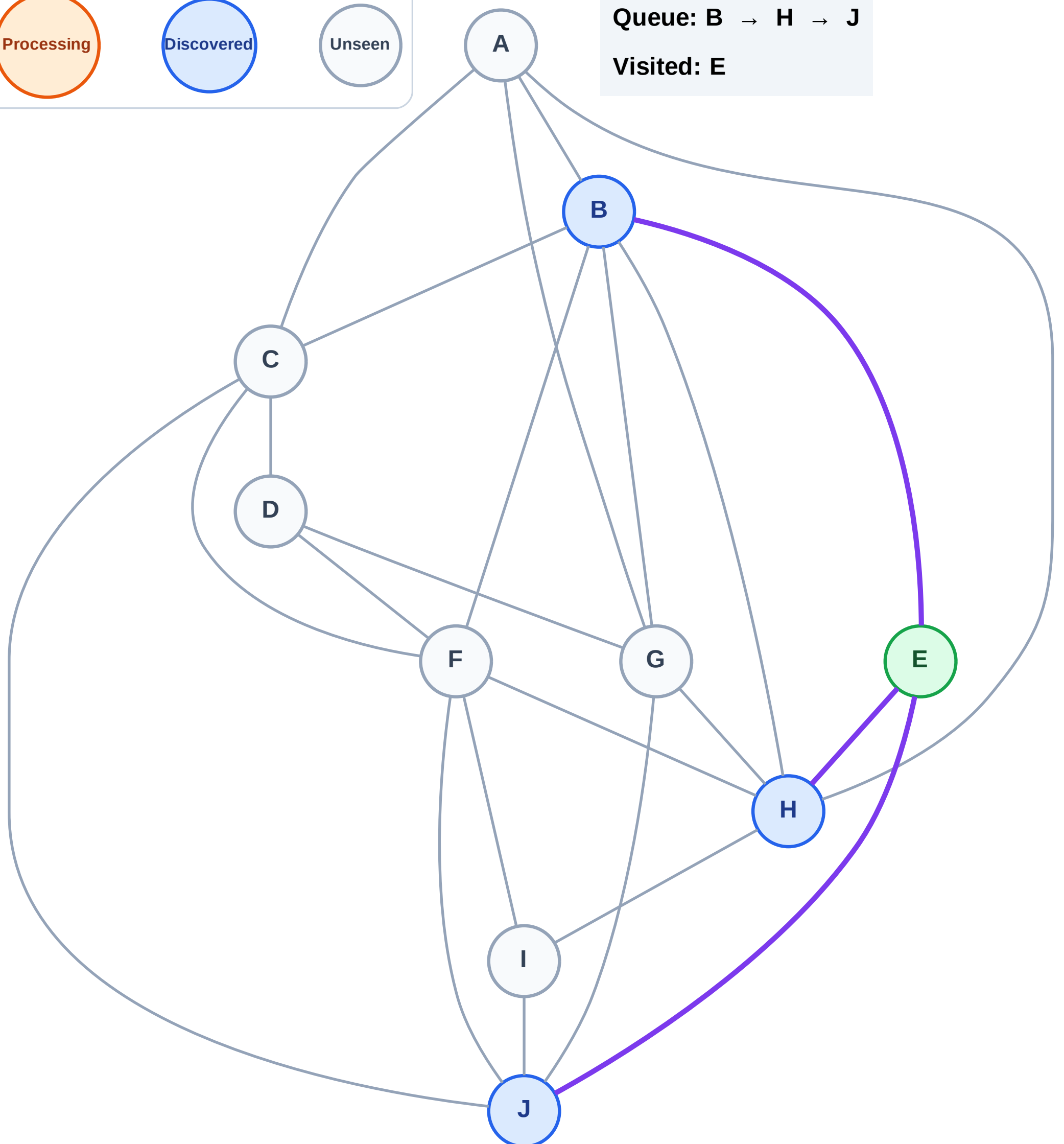
Step 3: Discover B, H, J from E

Legend



Queue: B → H → J

Visited: E



BFS Traversal

Step 4: Process node B

Legend

Complete

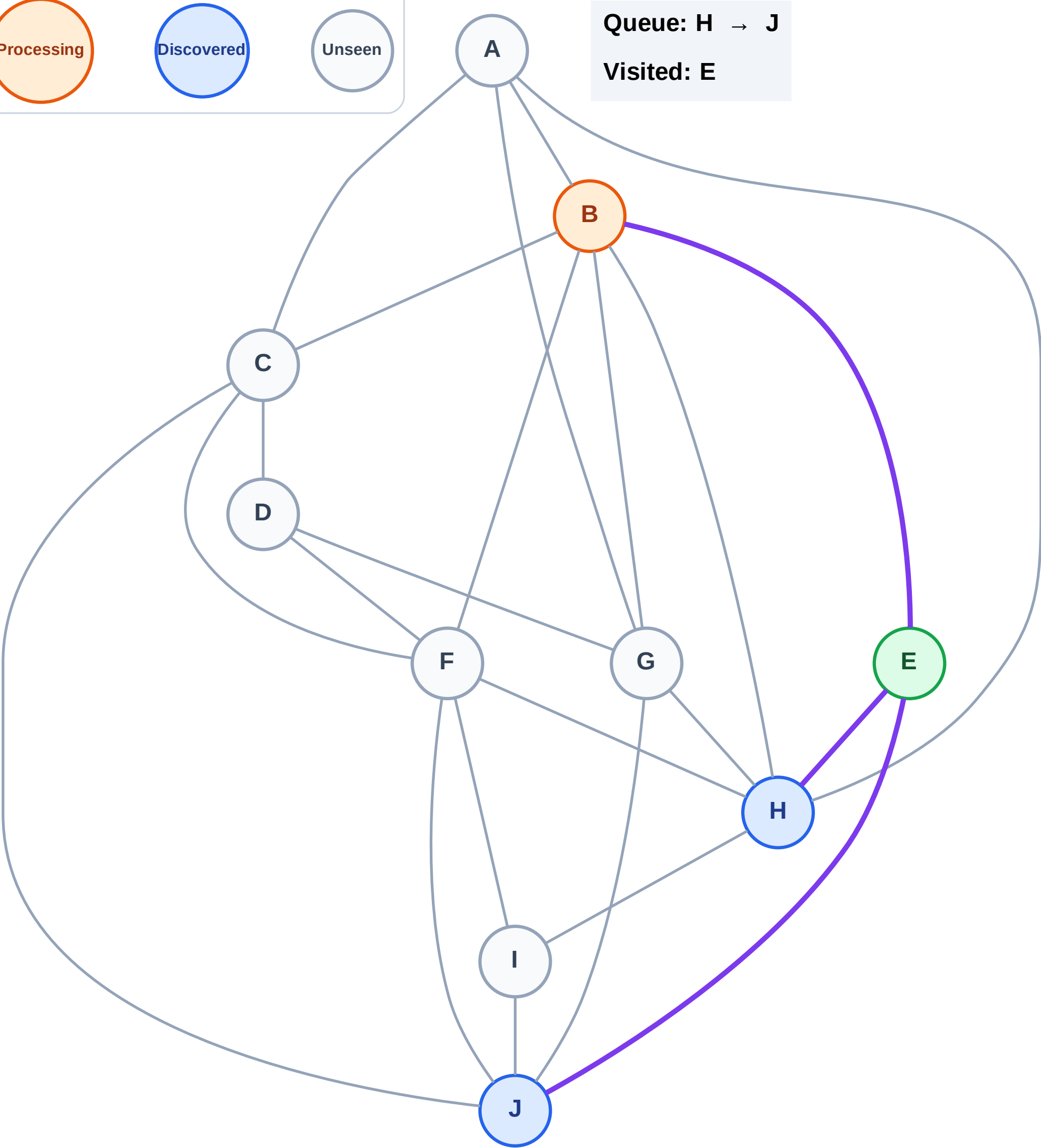
Processing

Discovered

Unseen

Queue: H → J

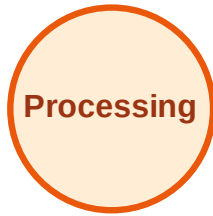
Visited: E



BFS Traversal

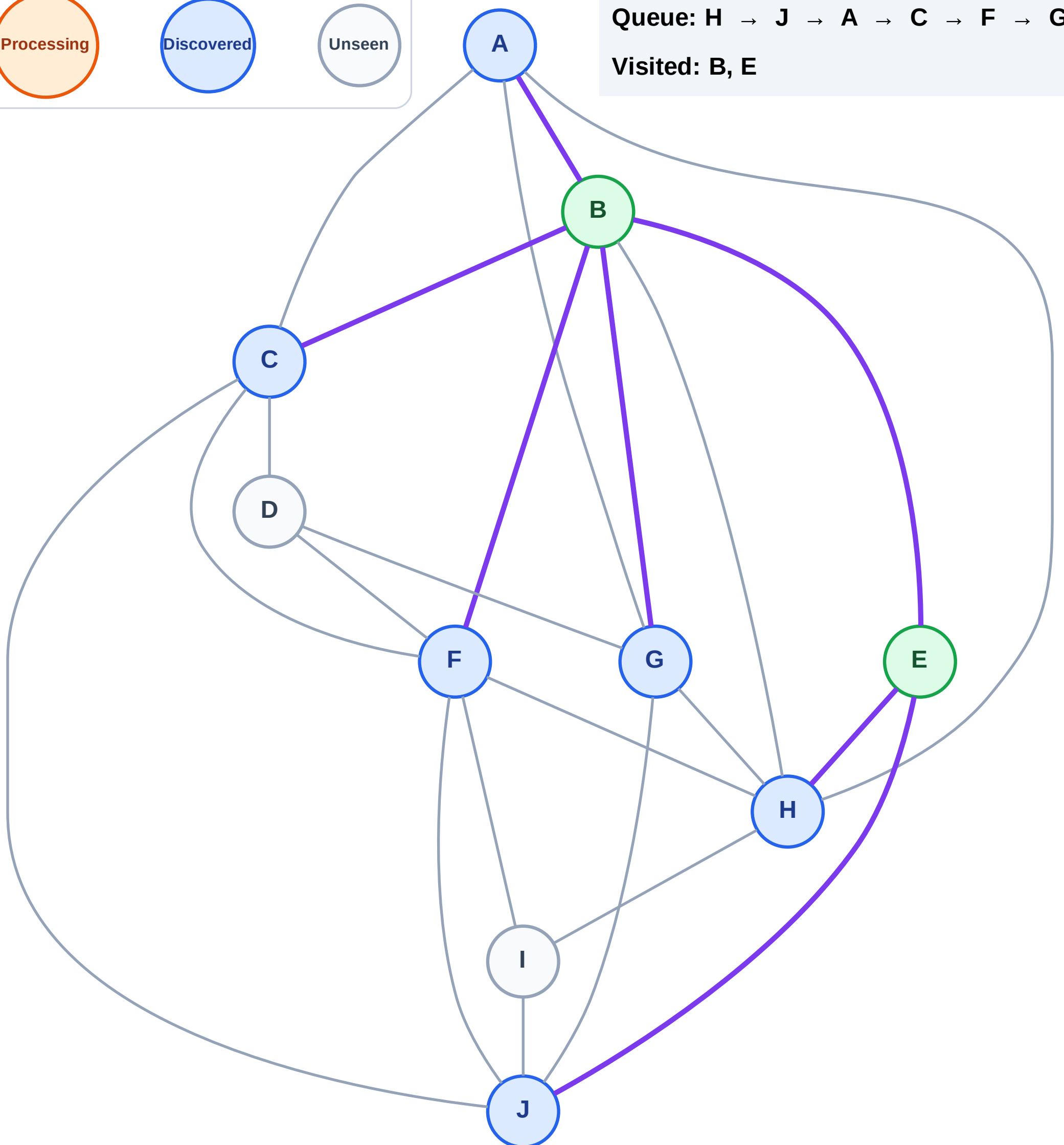
Step 5: Discover A, C, F, G from B

Legend



Queue: H → J → A → C → F → G

Visited: B, E



BFS Traversal

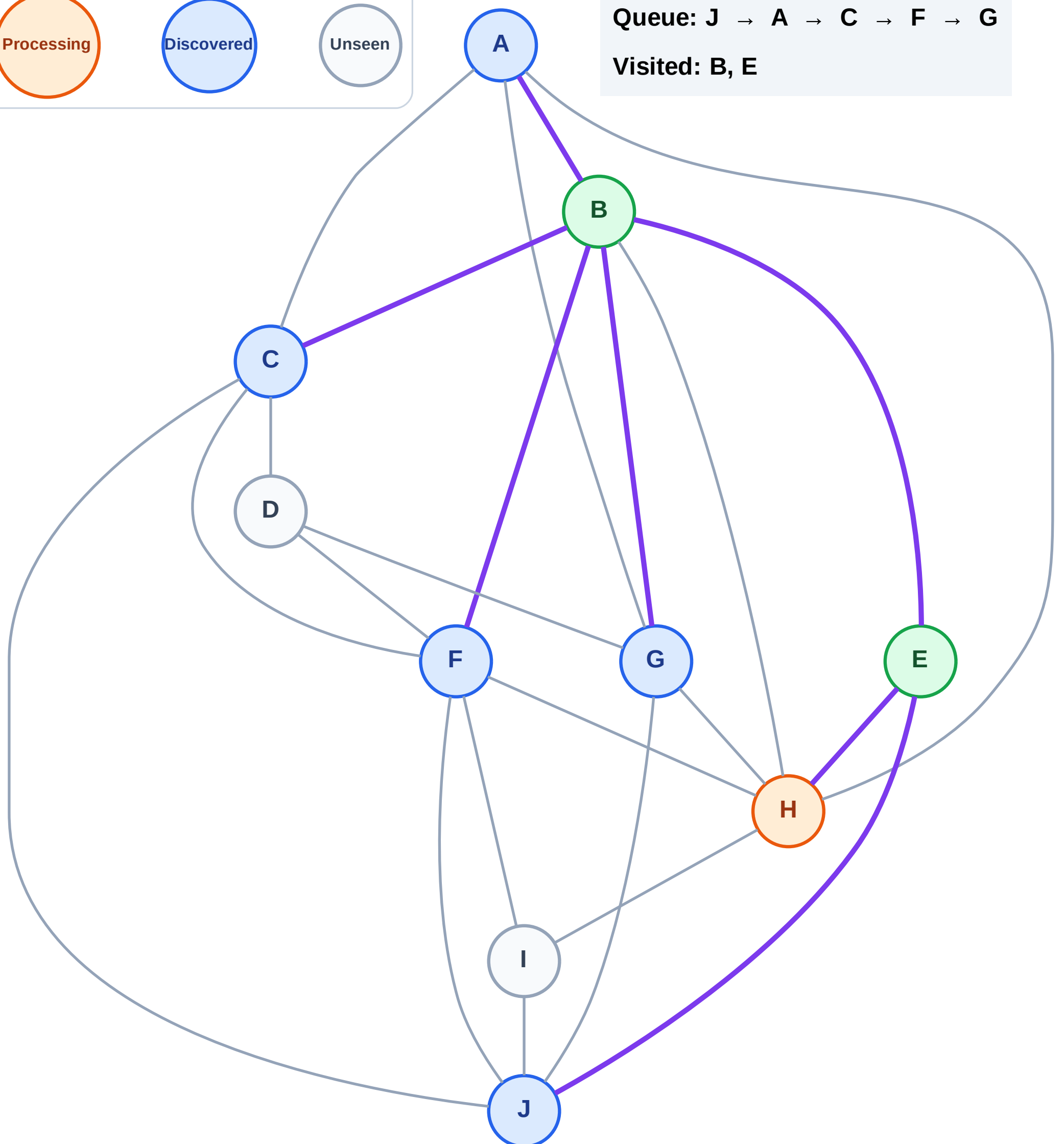
Step 6: Process node H

Legend



Queue: J → A → C → F → G

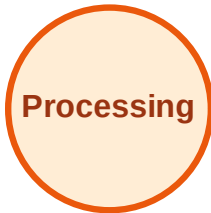
Visited: B, E



BFS Traversal

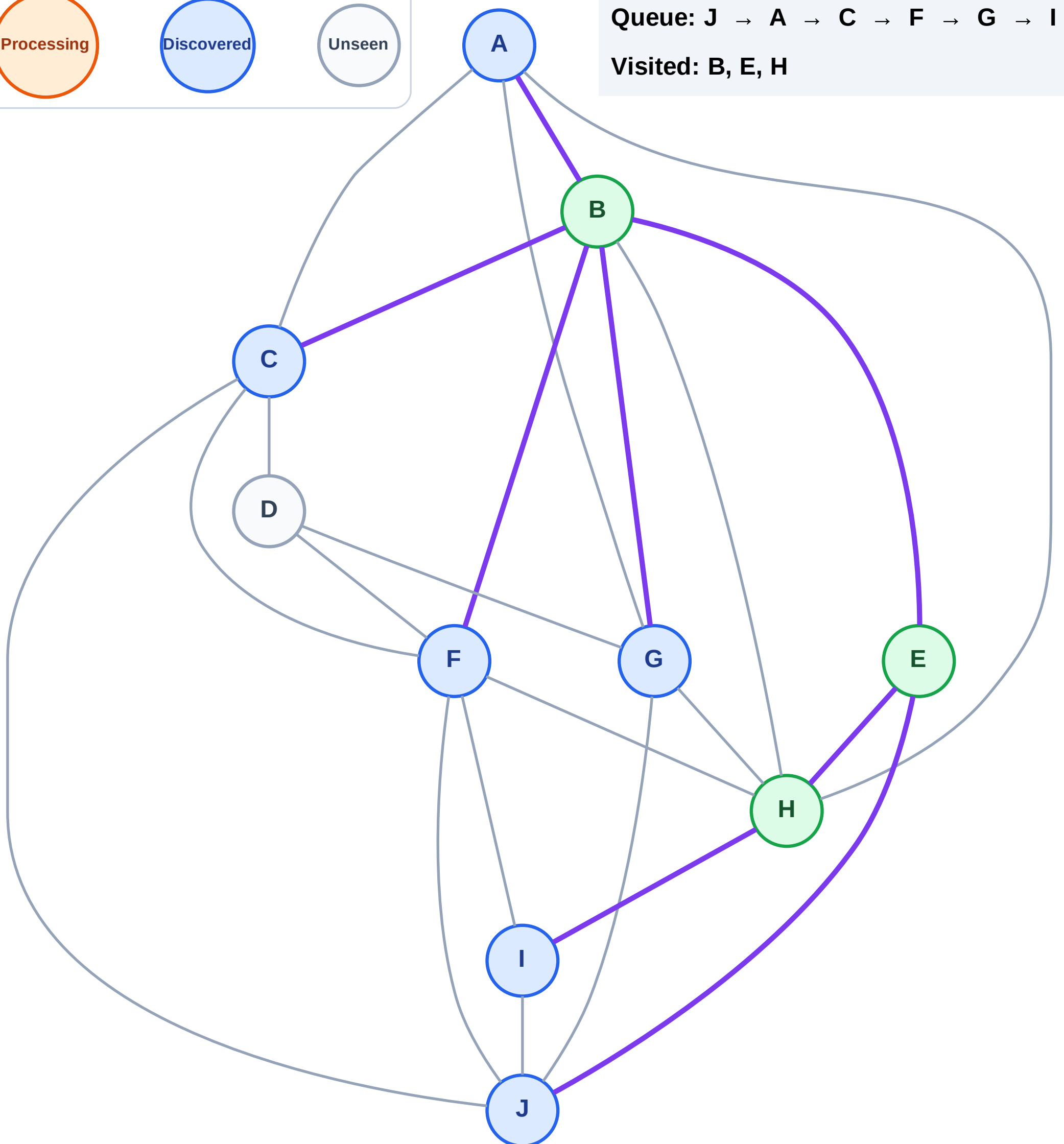
Step 7: Discover I from H

Legend



Queue: J → A → C → F → G → I

Visited: B, E, H



BFS Traversal

Step 8: Process node J

Complete

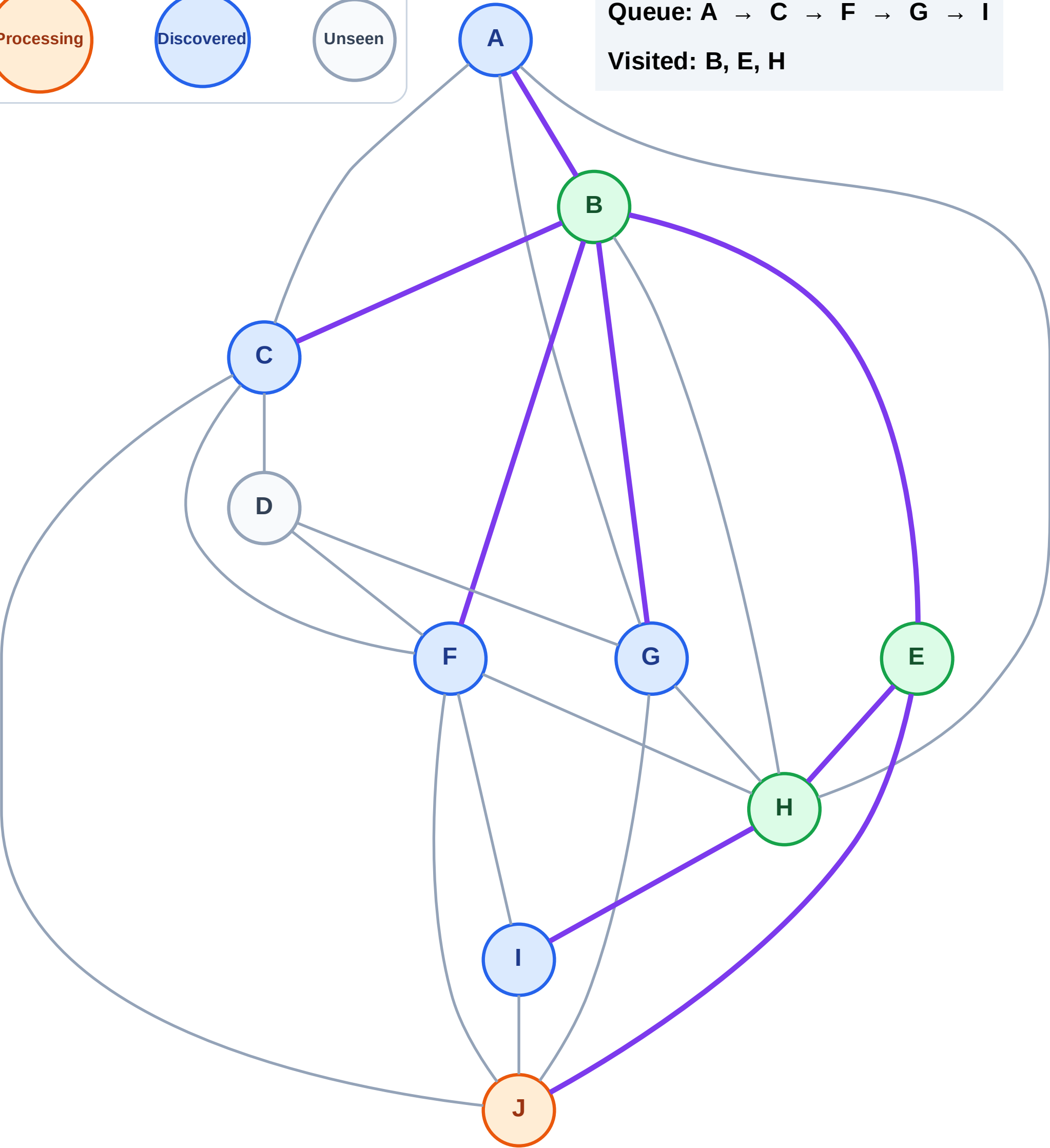
Processing

Discovered

Unseen

Queue: A → C → F → G → I

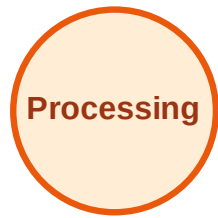
Visited: B, E, H



BFS Traversal

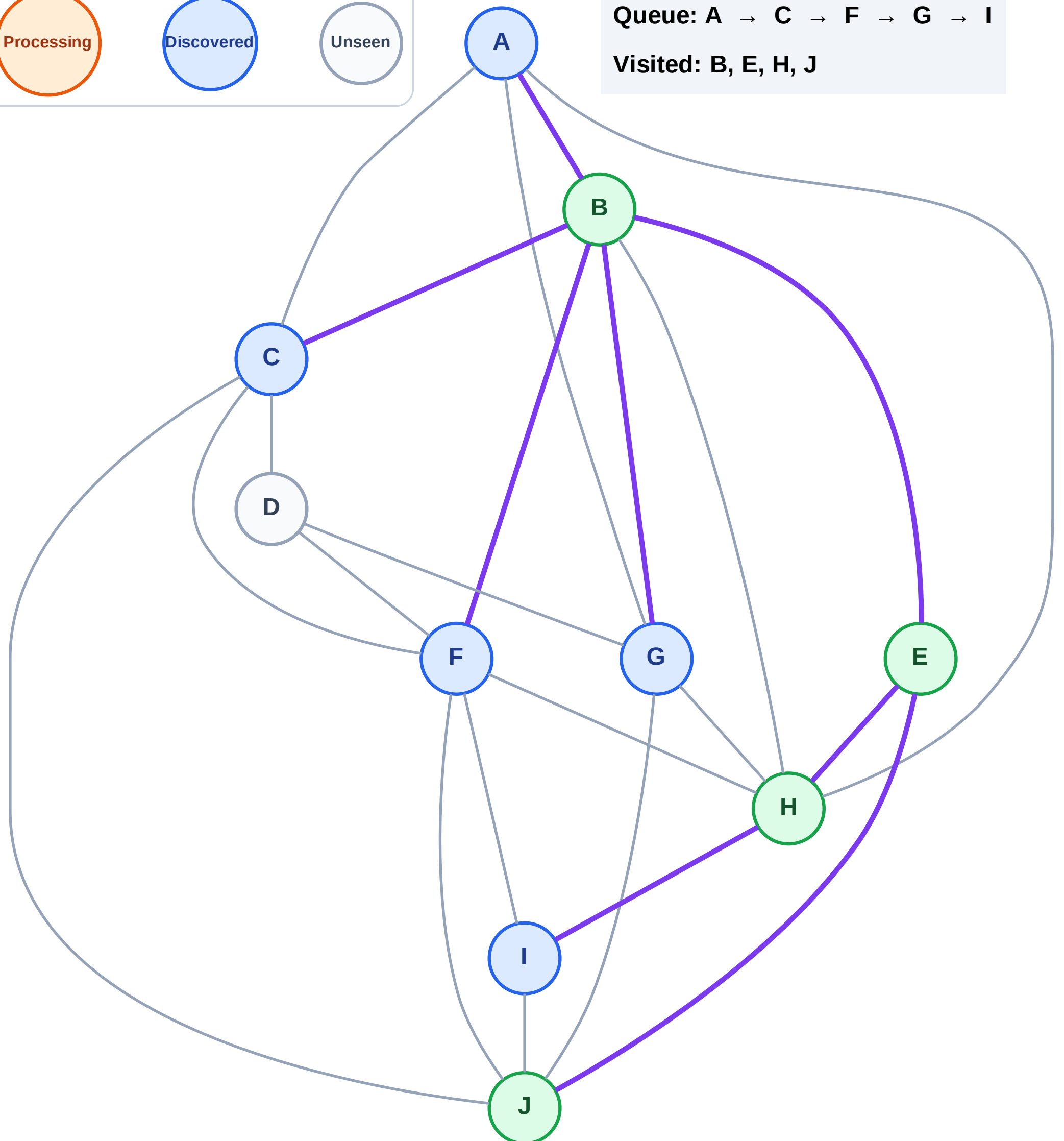
Step 9: No new neighbors from J

Legend



Queue: A → C → F → G → I

Visited: B, E, H, J



BFS Traversal

Step 10: Process node A

Legend

Complete

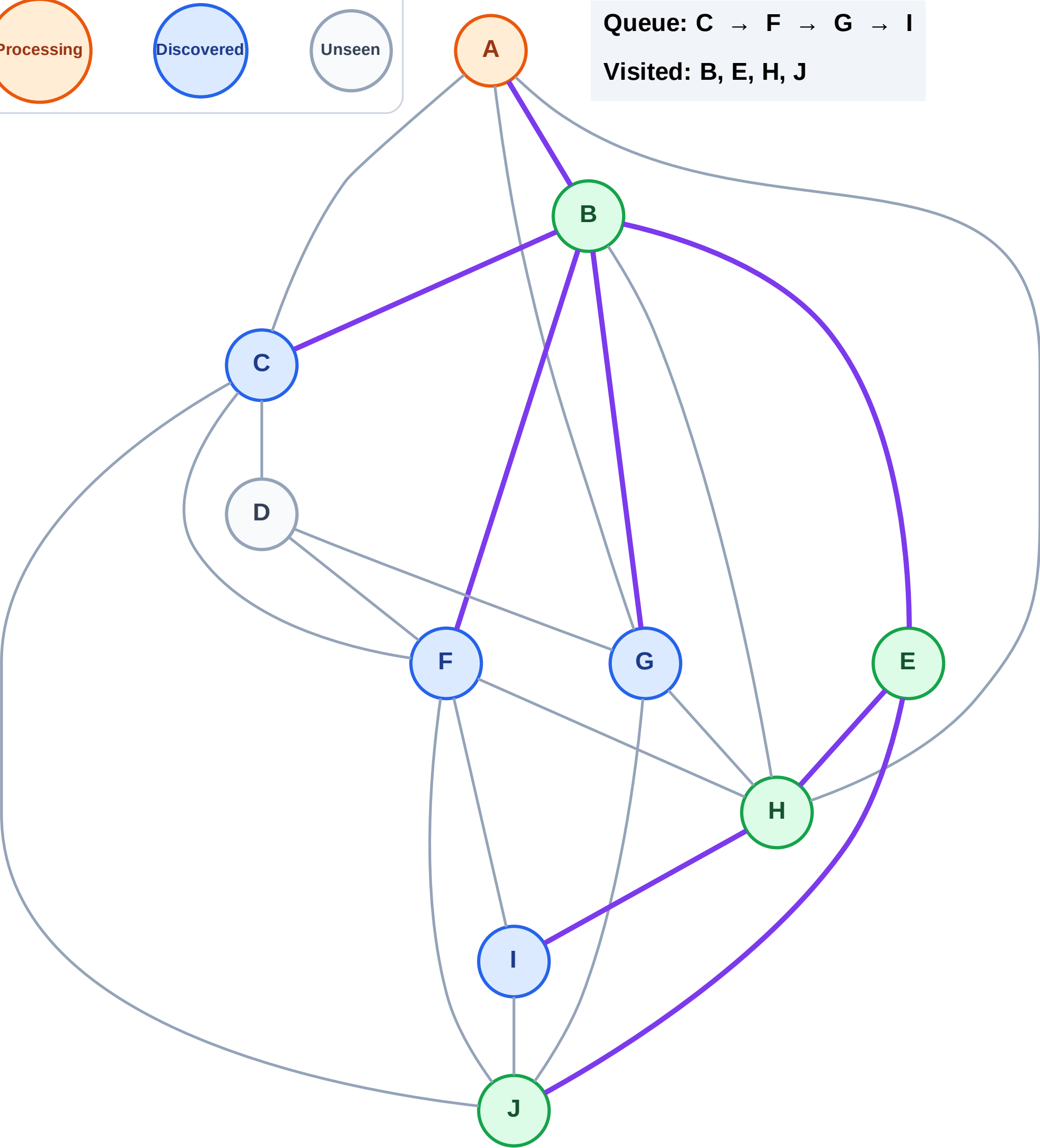
Processing

Discovered

Unseen

Queue: C → F → G → I

Visited: B, E, H, J



BFS Traversal

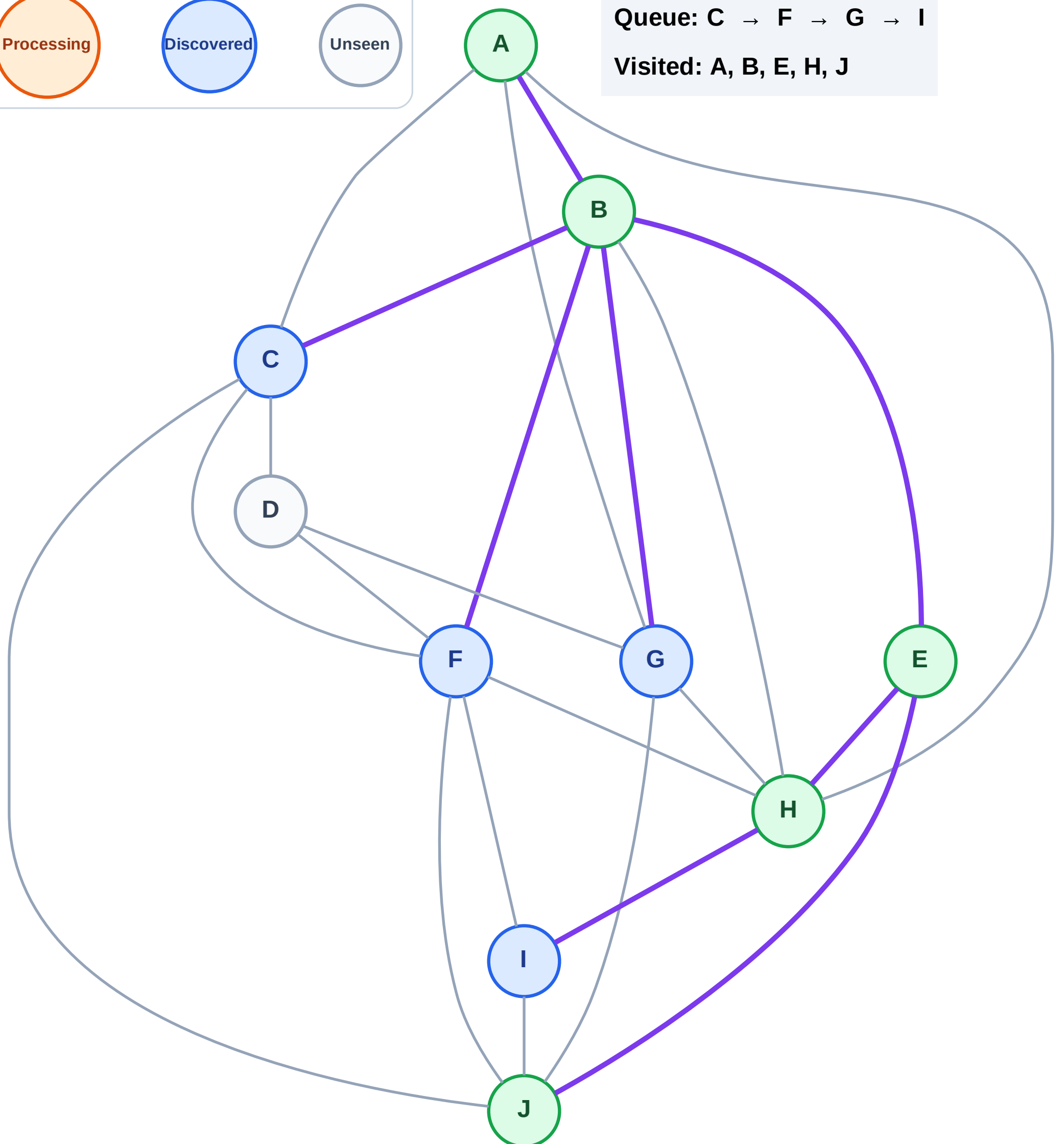
Step 11: No new neighbors from A

Legend



Queue: C → F → G → I

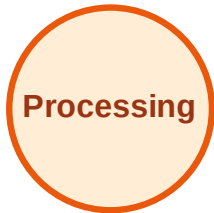
Visited: A, B, E, H, J



BFS Traversal

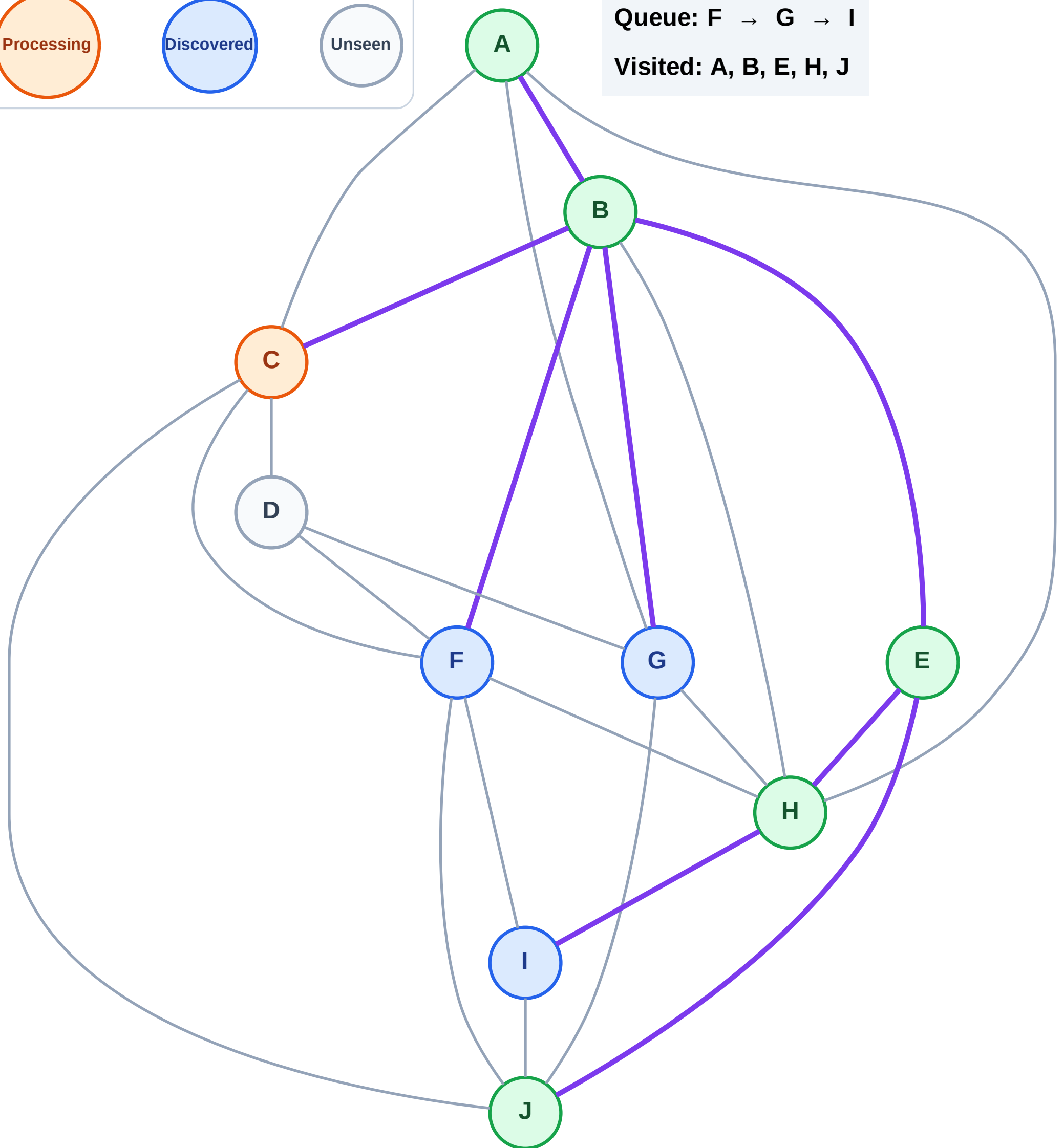
Step 12: Process node C

Legend



Queue: F → G → I

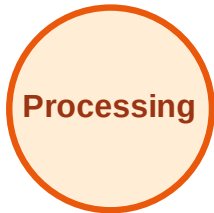
Visited: A, B, E, H, J



BFS Traversal

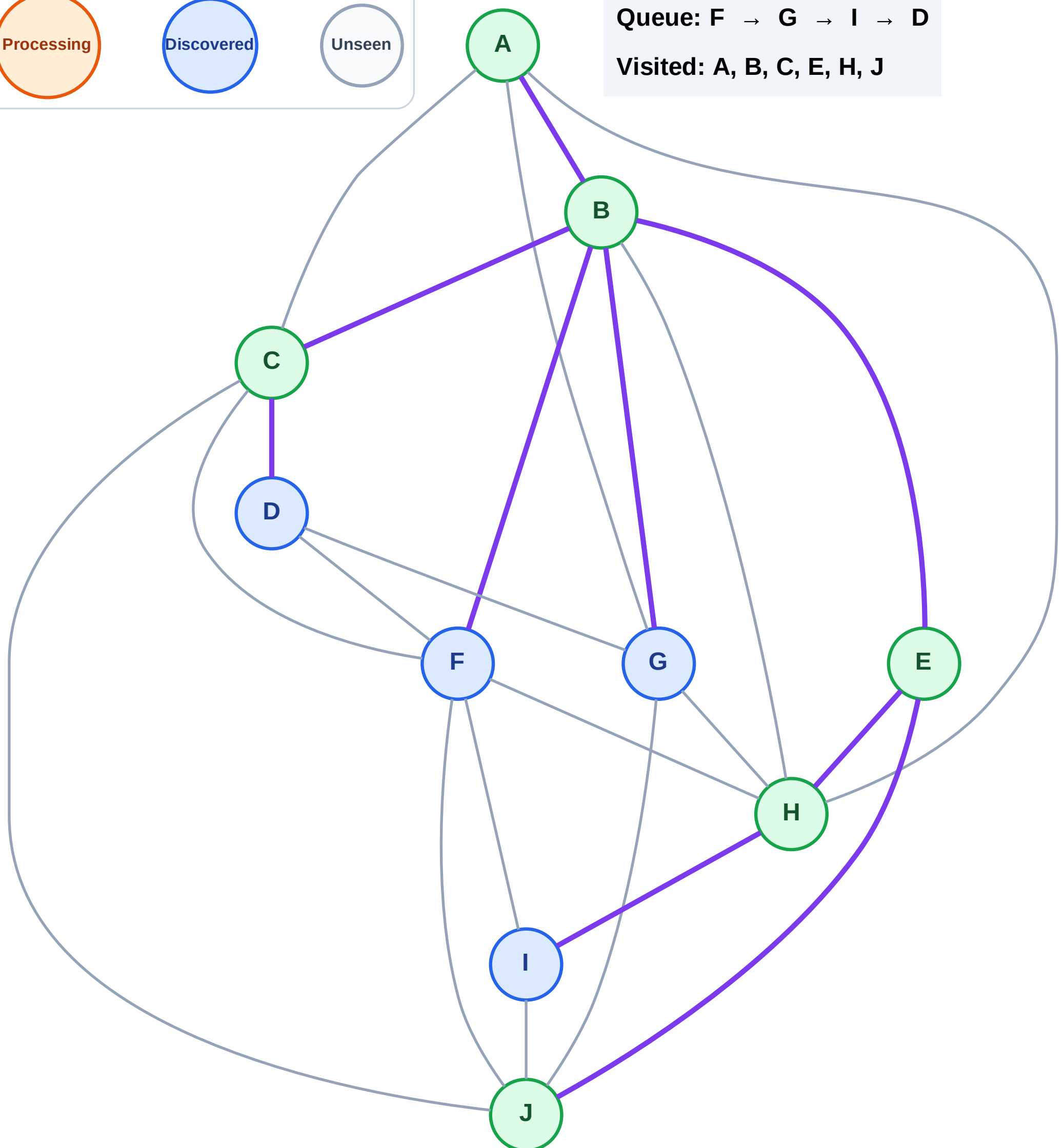
Step 13: Discover D from C

Legend



Queue: F → G → I → D

Visited: A, B, C, E, H, J



BFS Traversal

Step 14: Process node F

Legend

Complete

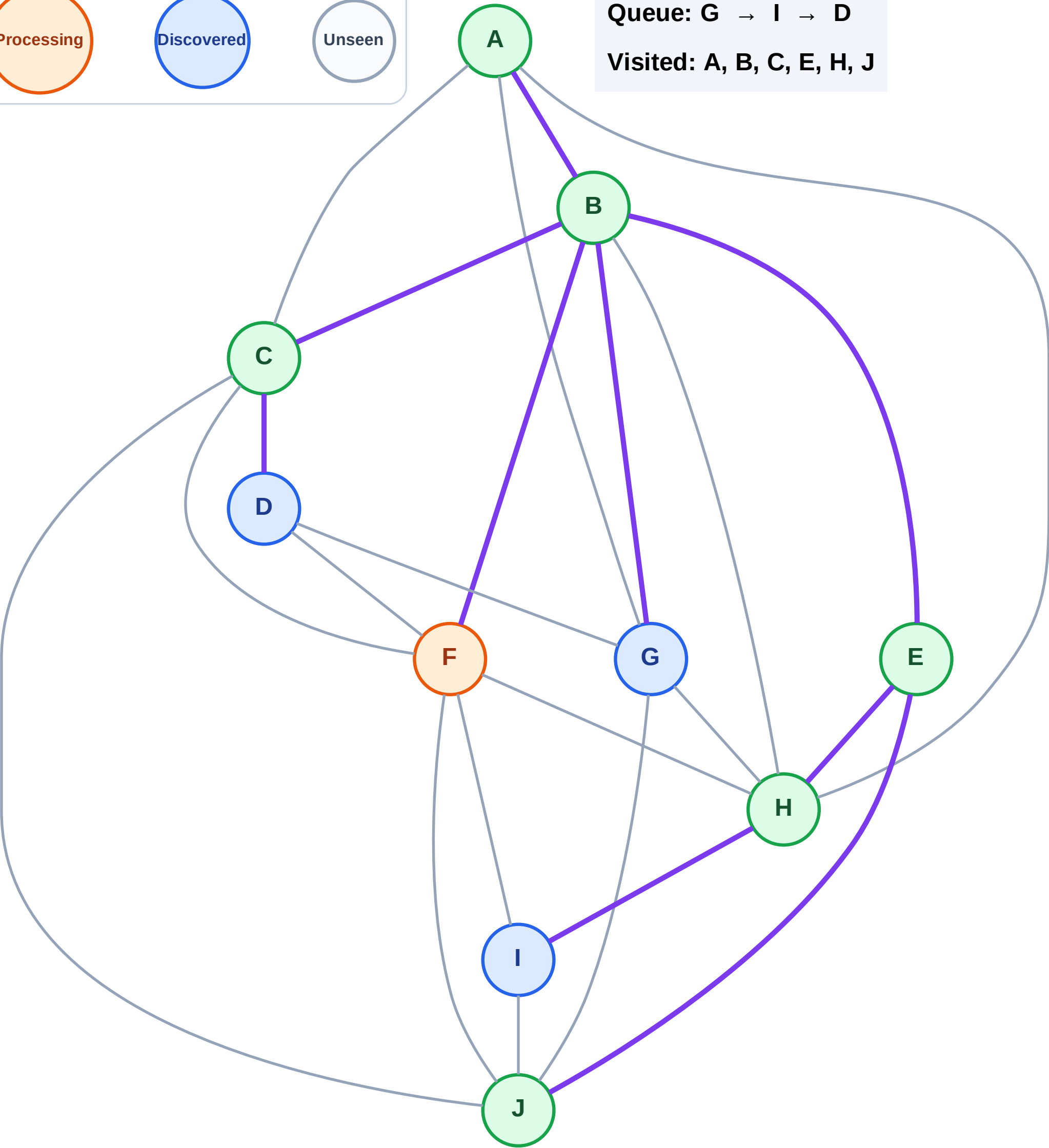
Processing

Discovered

Unseen

Queue: G → I → D

Visited: A, B, C, E, H, J



BFS Traversal

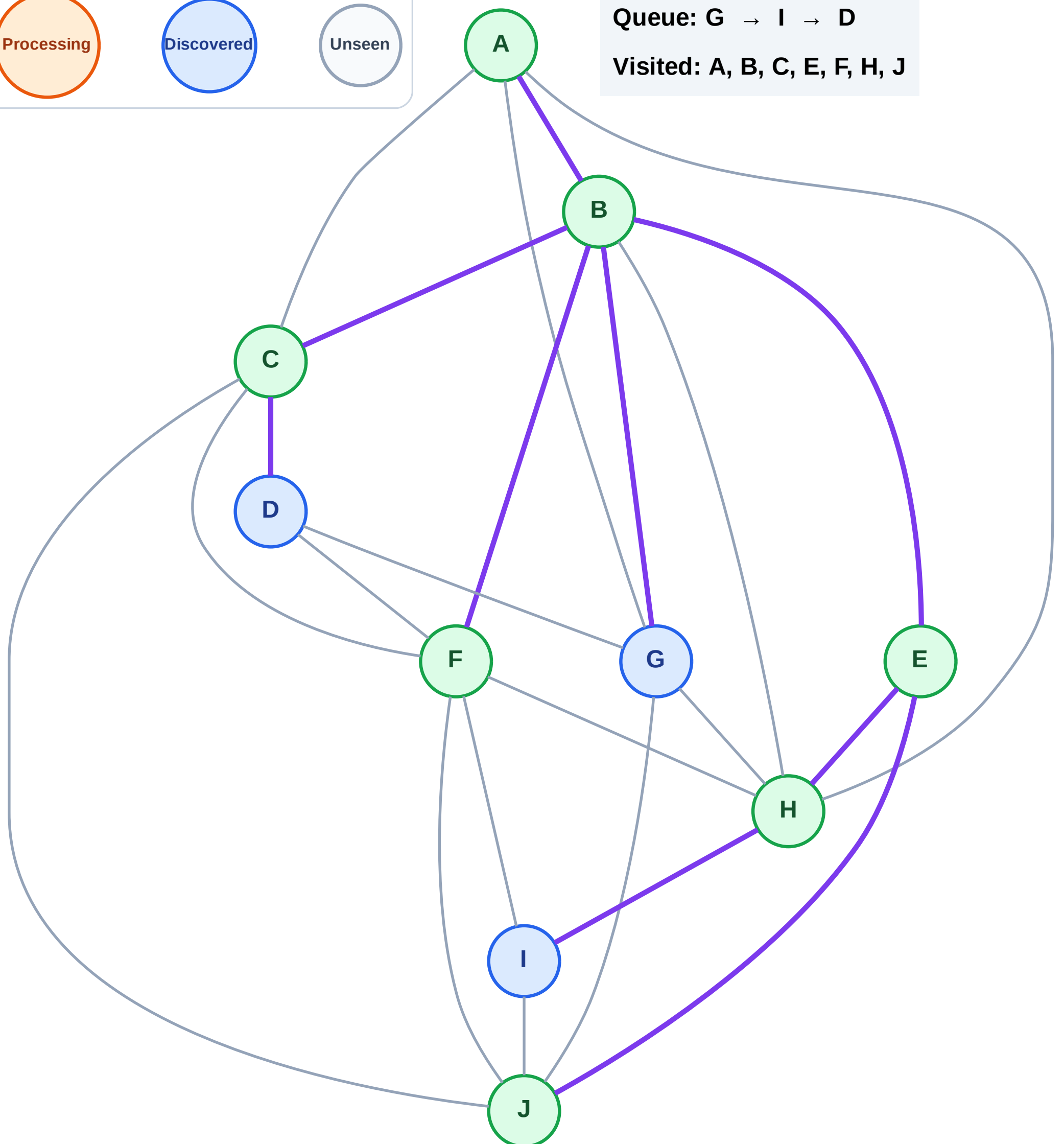
Step 15: No new neighbors from F

Legend



Queue: G → I → D

Visited: A, B, C, E, F, H, J



BFS Traversal

Step 16: Process node G

Legend

Complete

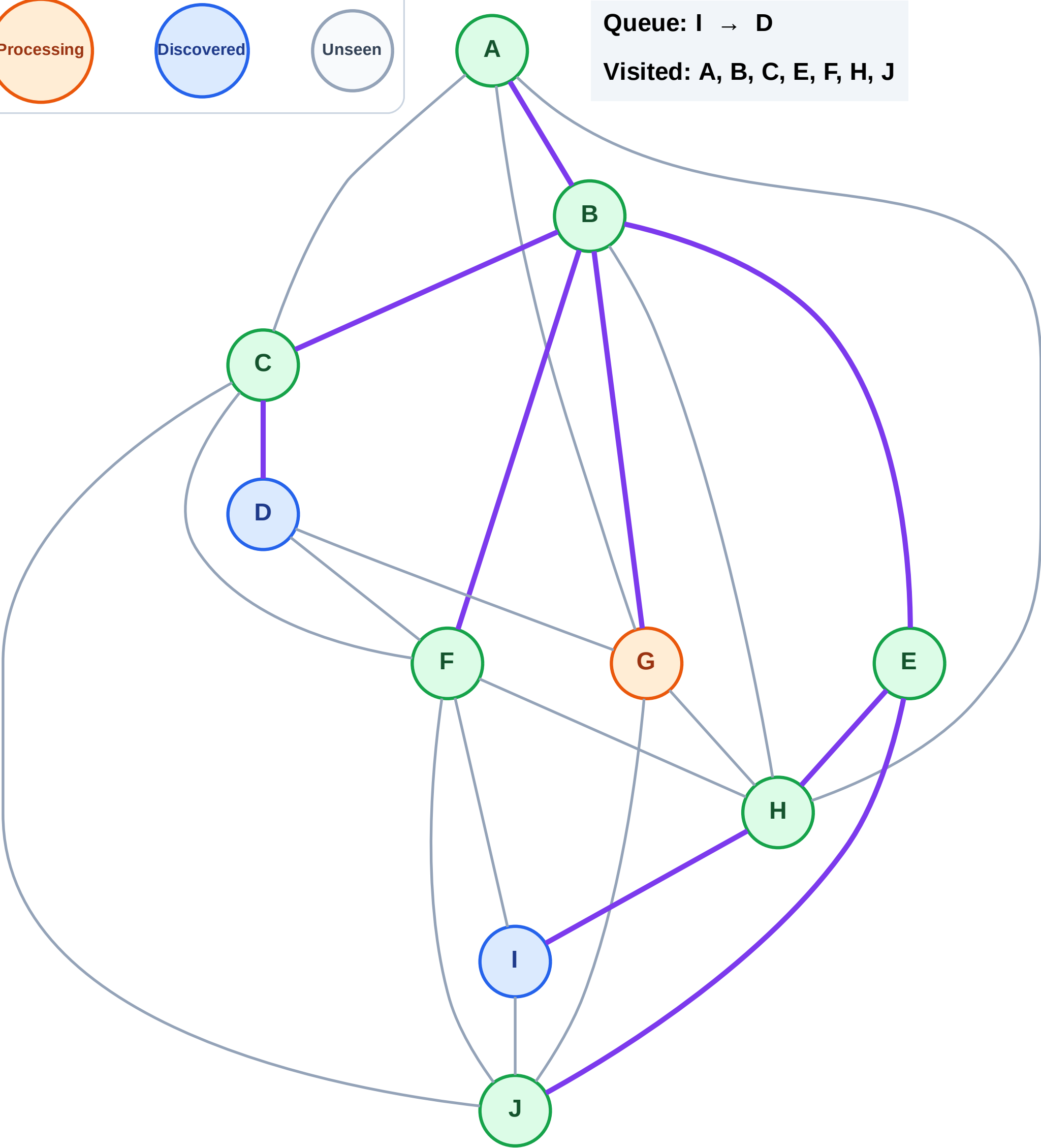
Processing

Discovered

Unseen

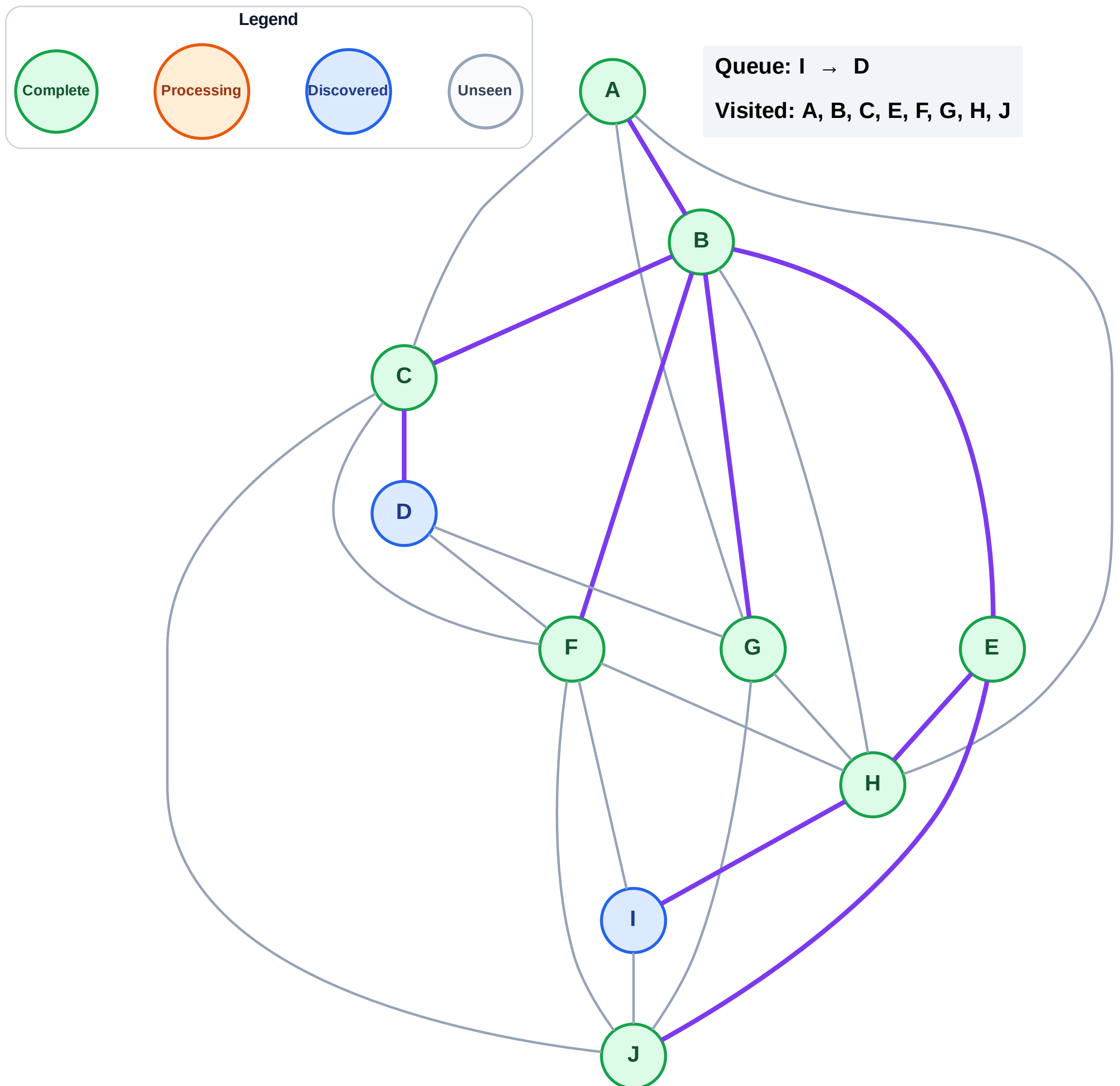
Queue: I → D

Visited: A, B, C, E, F, H, J



Step 17: No new neighbors from G

Step 17: No new neighbors from G



BFS Traversal

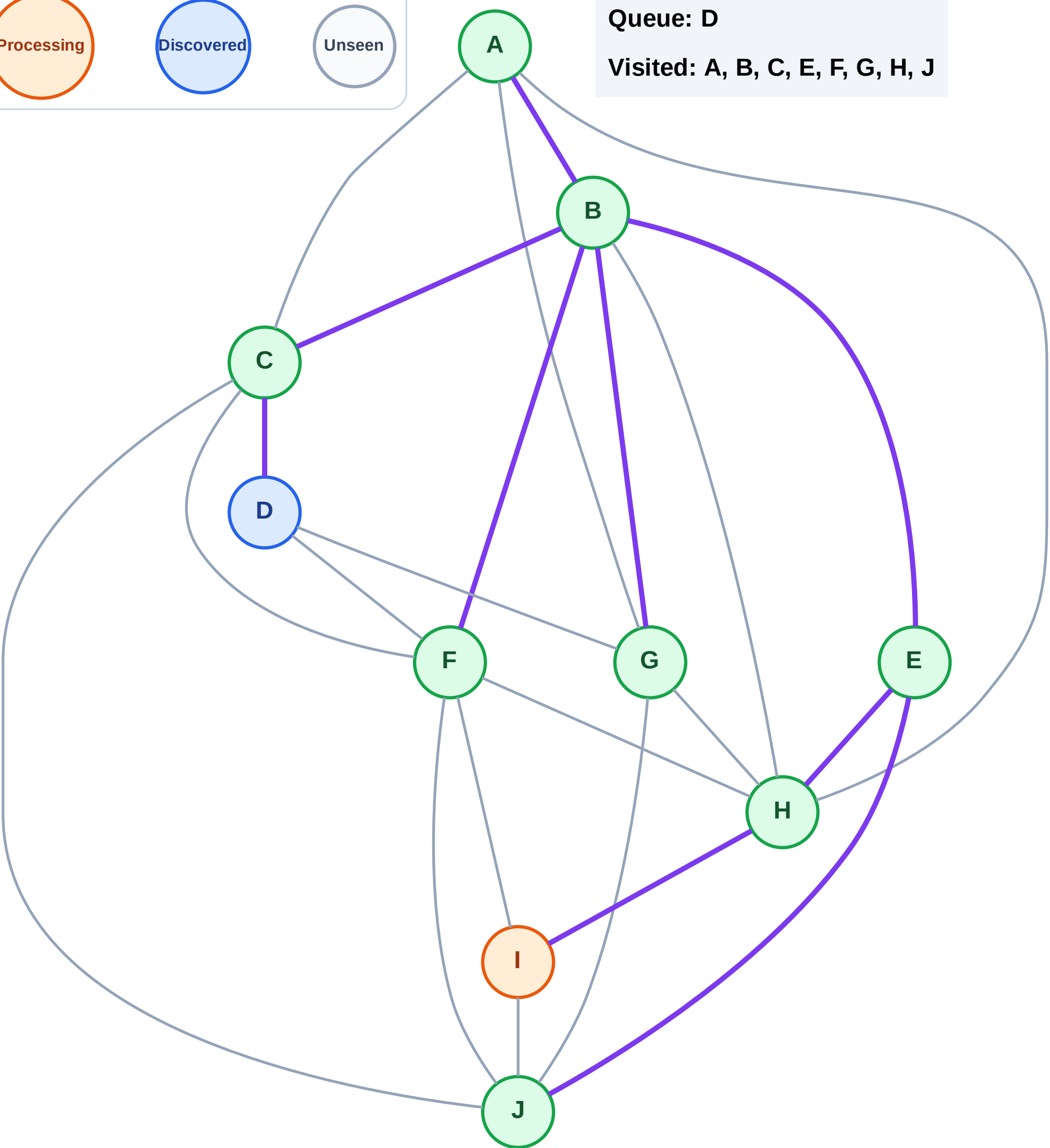
Step 18: Process node I

Legend



Queue: D

Visited: A, B, C, E, F, G, H, J



BFS Traversal

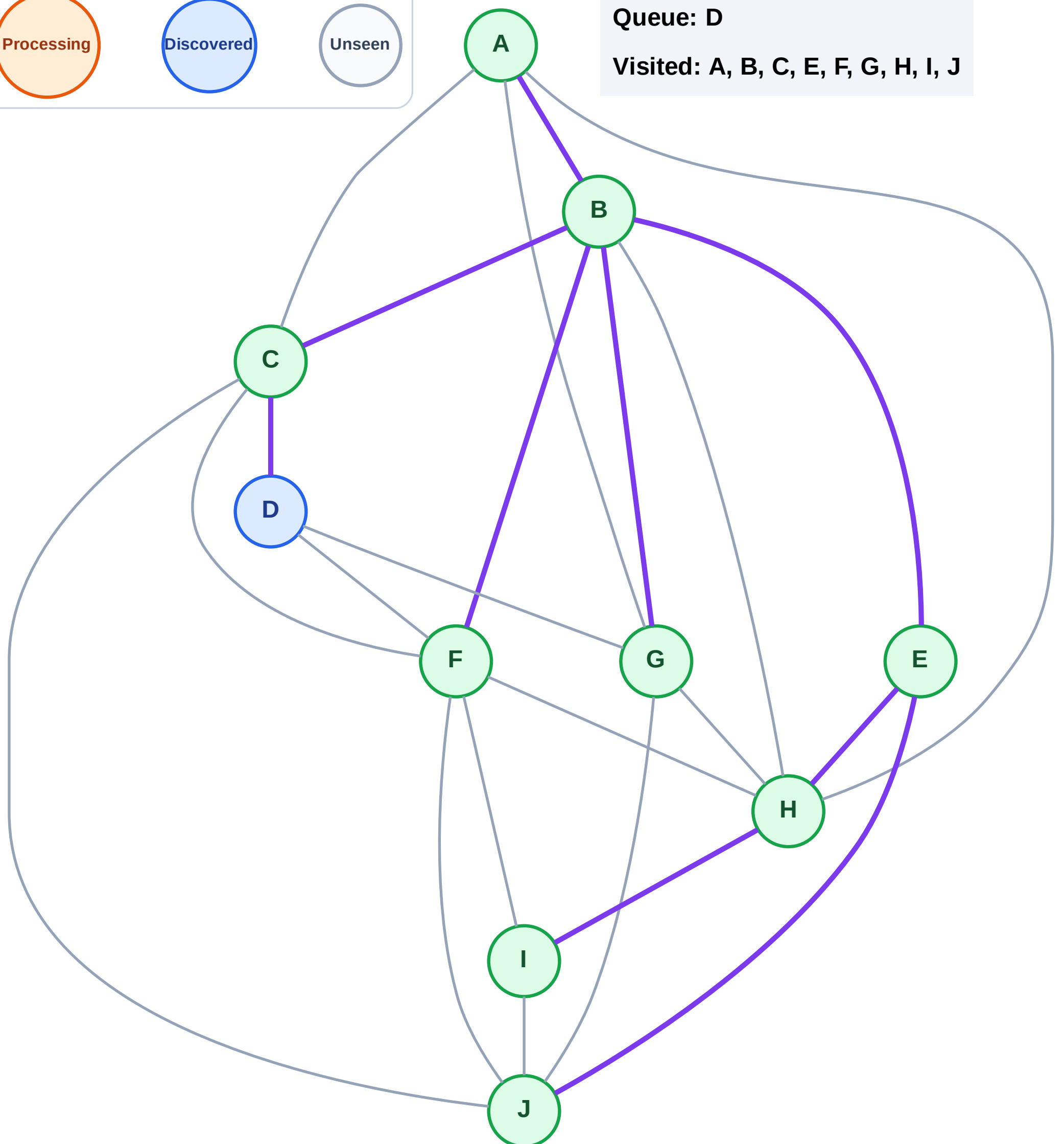
Step 19: No new neighbors from I

Legend



Queue: D

Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

Step 20: Process node D

Legend

Complete

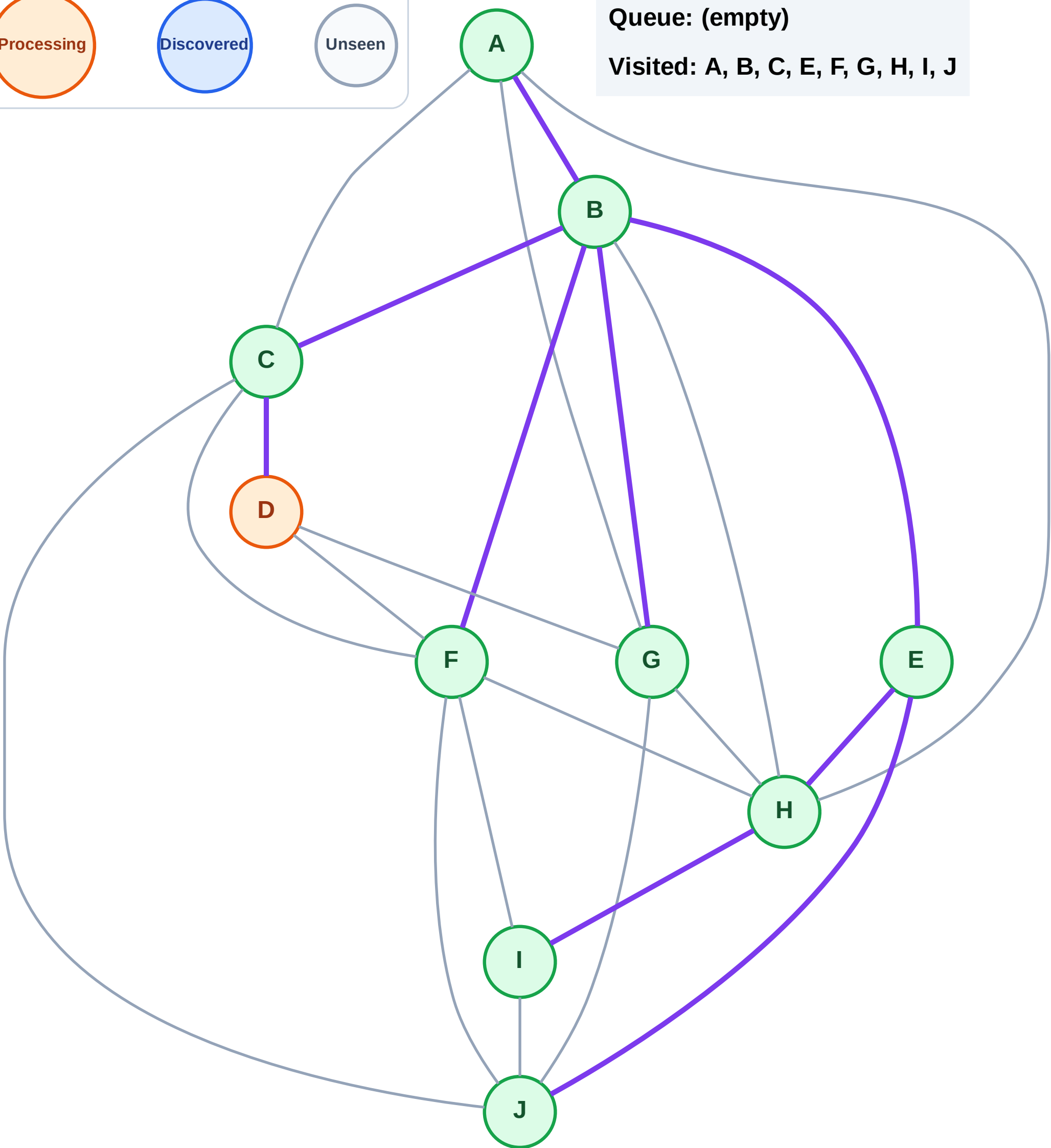
Processing

Discovered

Unseen

Queue: (empty)

Visited: A, B, C, E, F, G, H, I, J



BFS Traversal

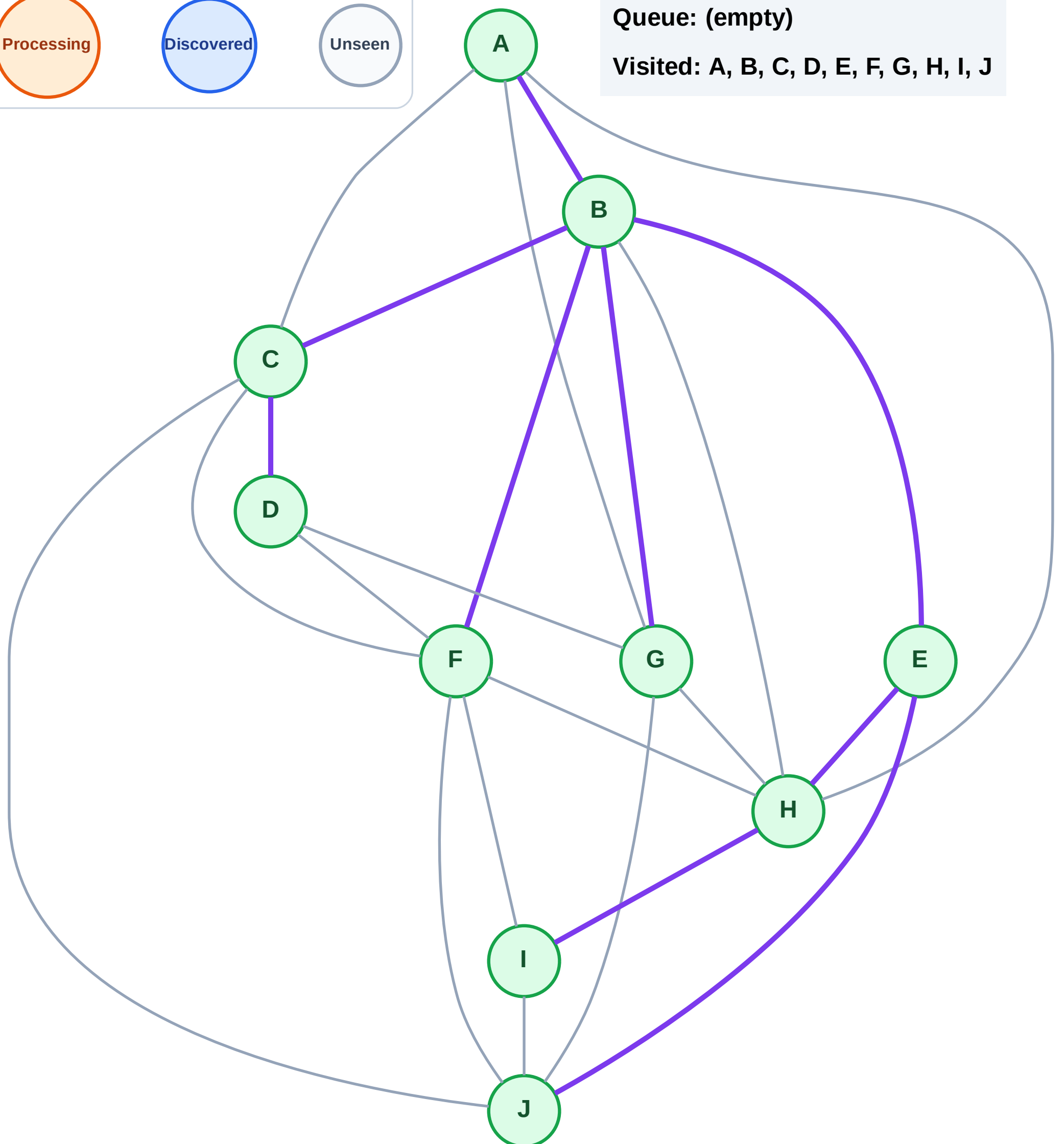
Step 21: No new neighbors from D

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

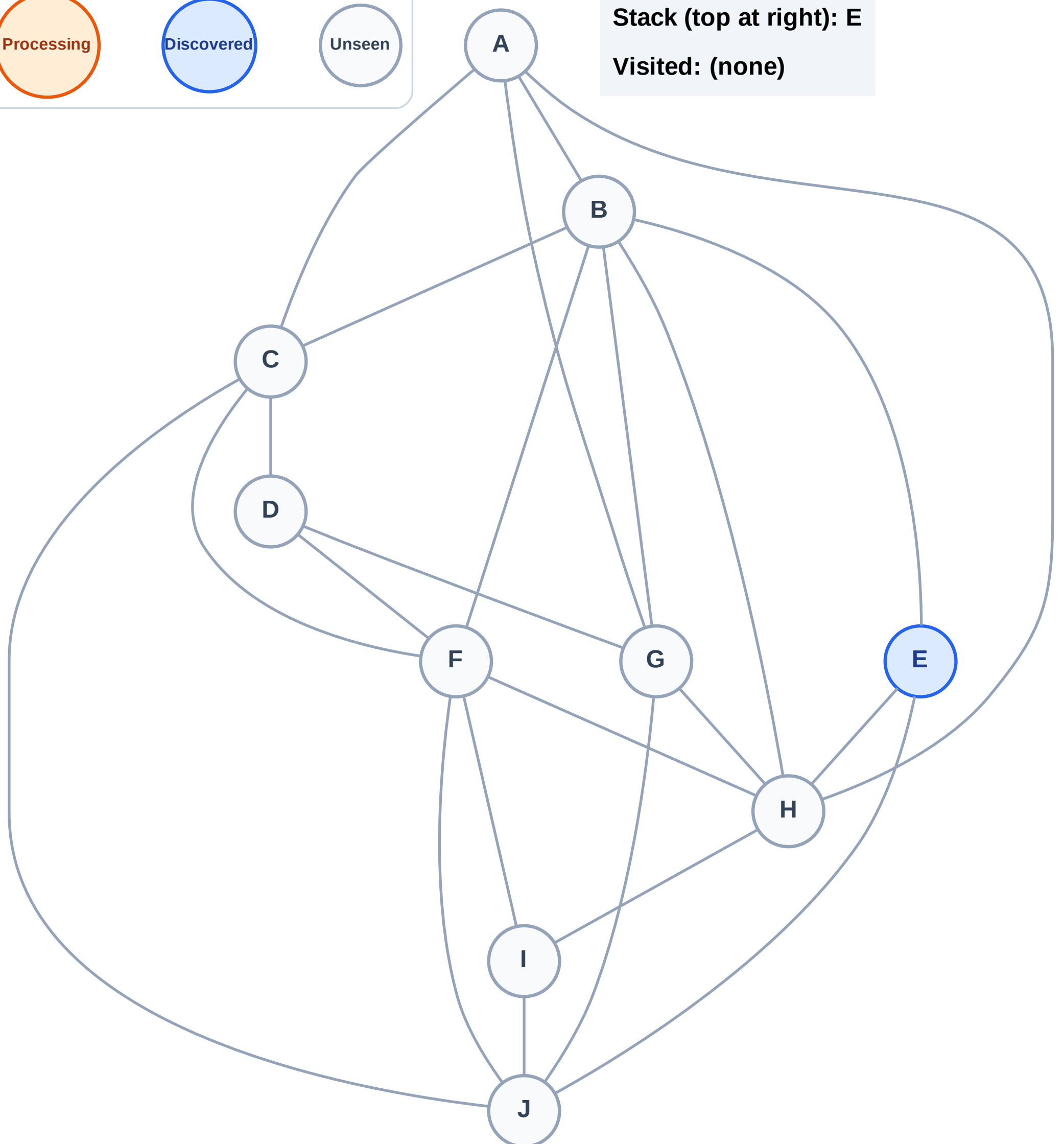
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

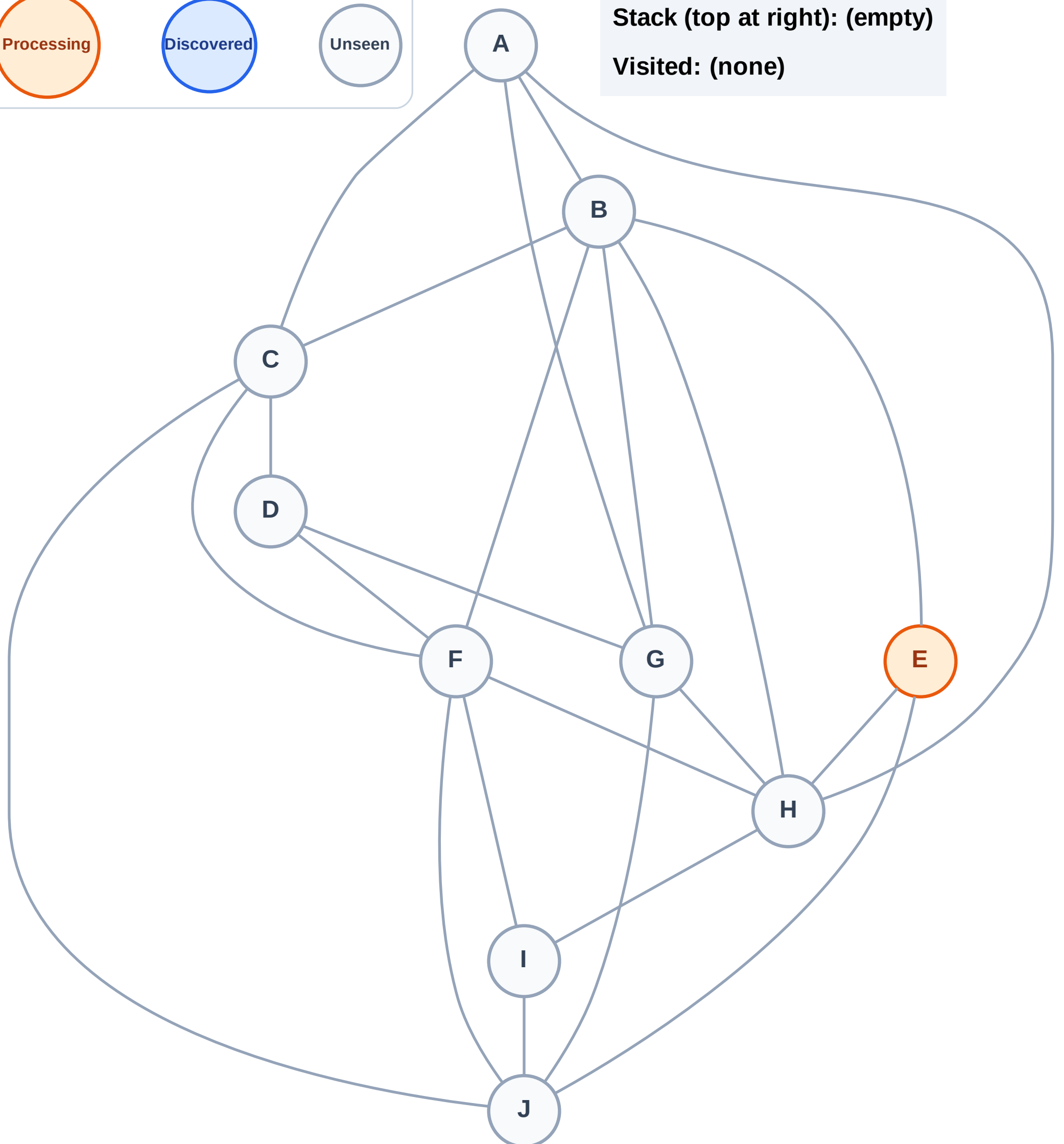
Step 2: Process node E

Legend



Stack (top at right): (empty)

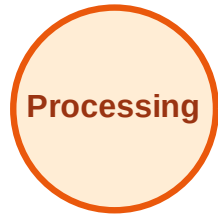
Visited: (none)



DFS Traversal

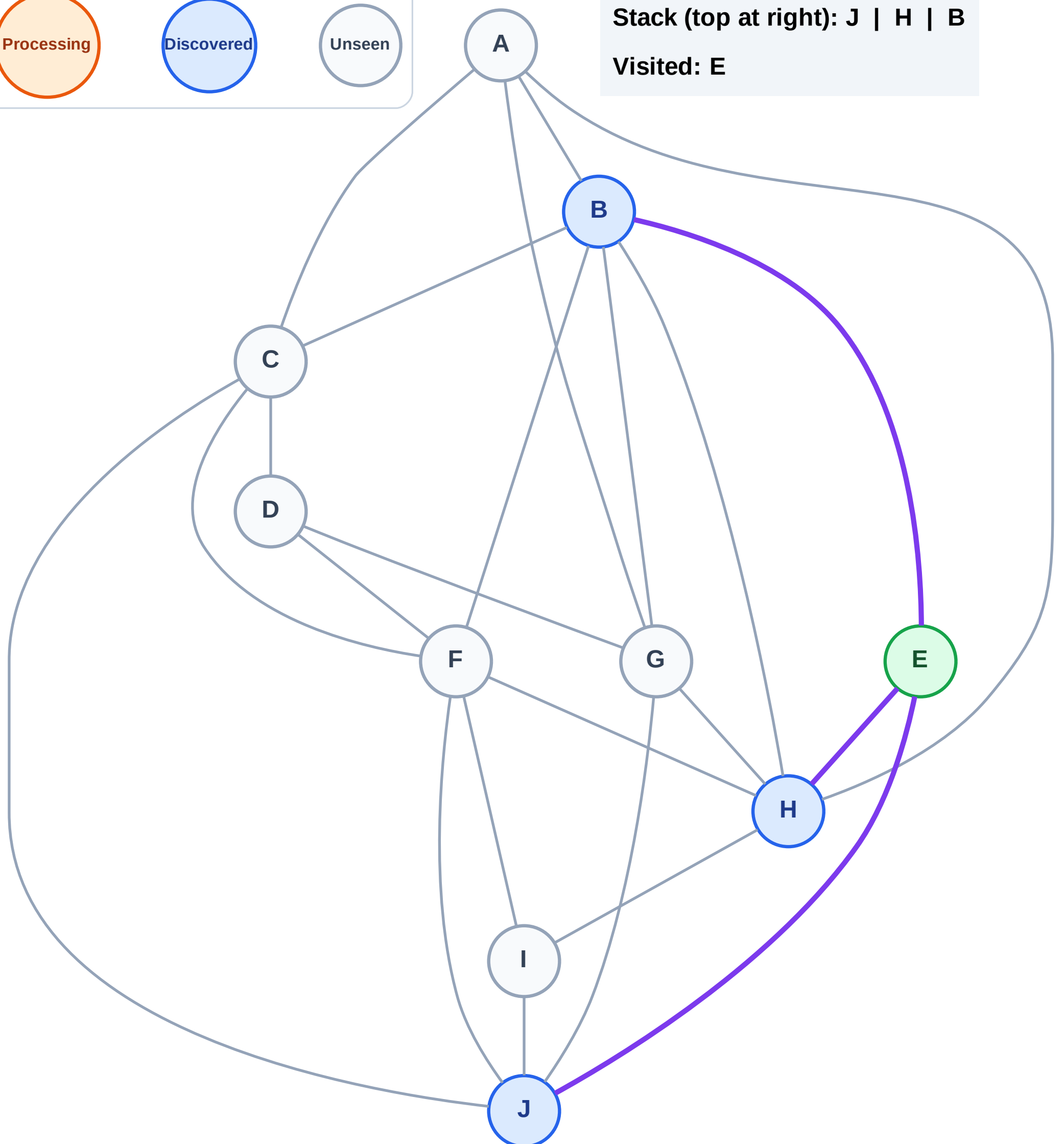
Step 3: Discover J, H, B from E

Legend



Stack (top at right): J | H | B

Visited: E



DFS Traversal

Step 4: Process node B

Legend

Complete

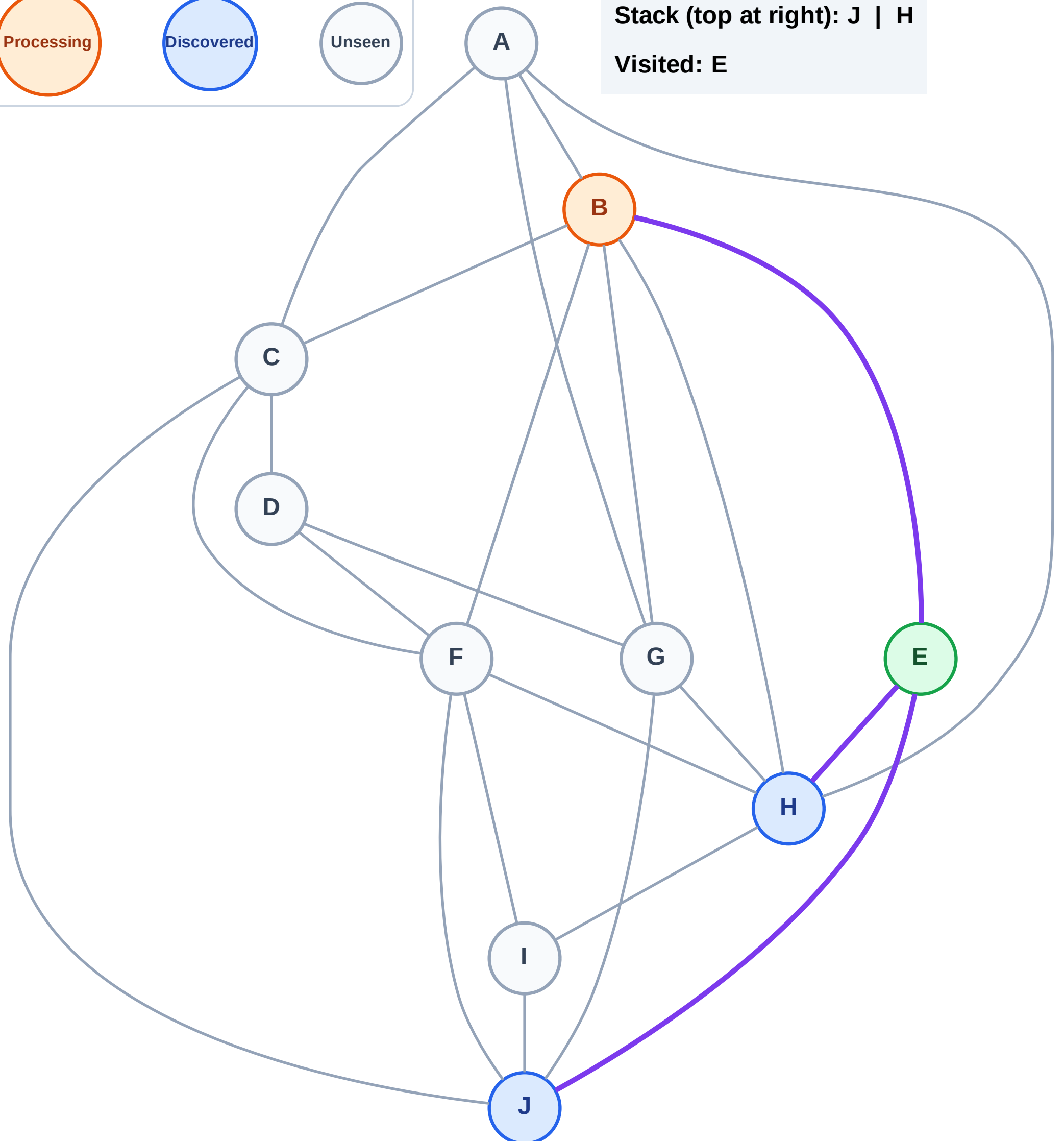
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: E



DFS Traversal

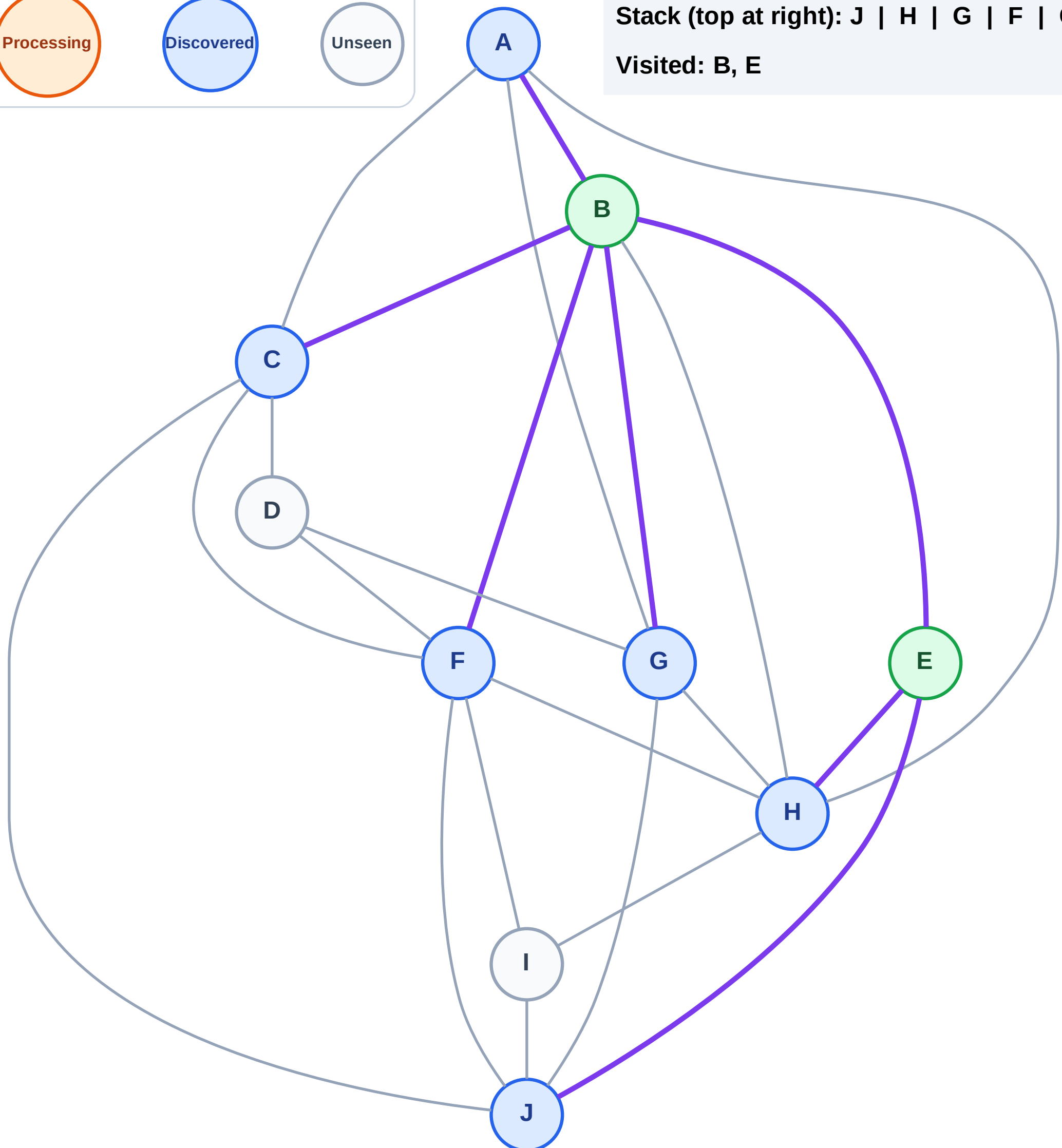
Step 5: Discover G, F, C, A from B

Legend



Stack (top at right): J | H | G | F | C | A

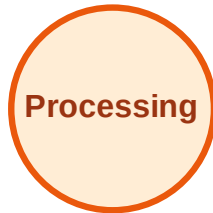
Visited: B, E



DFS Traversal

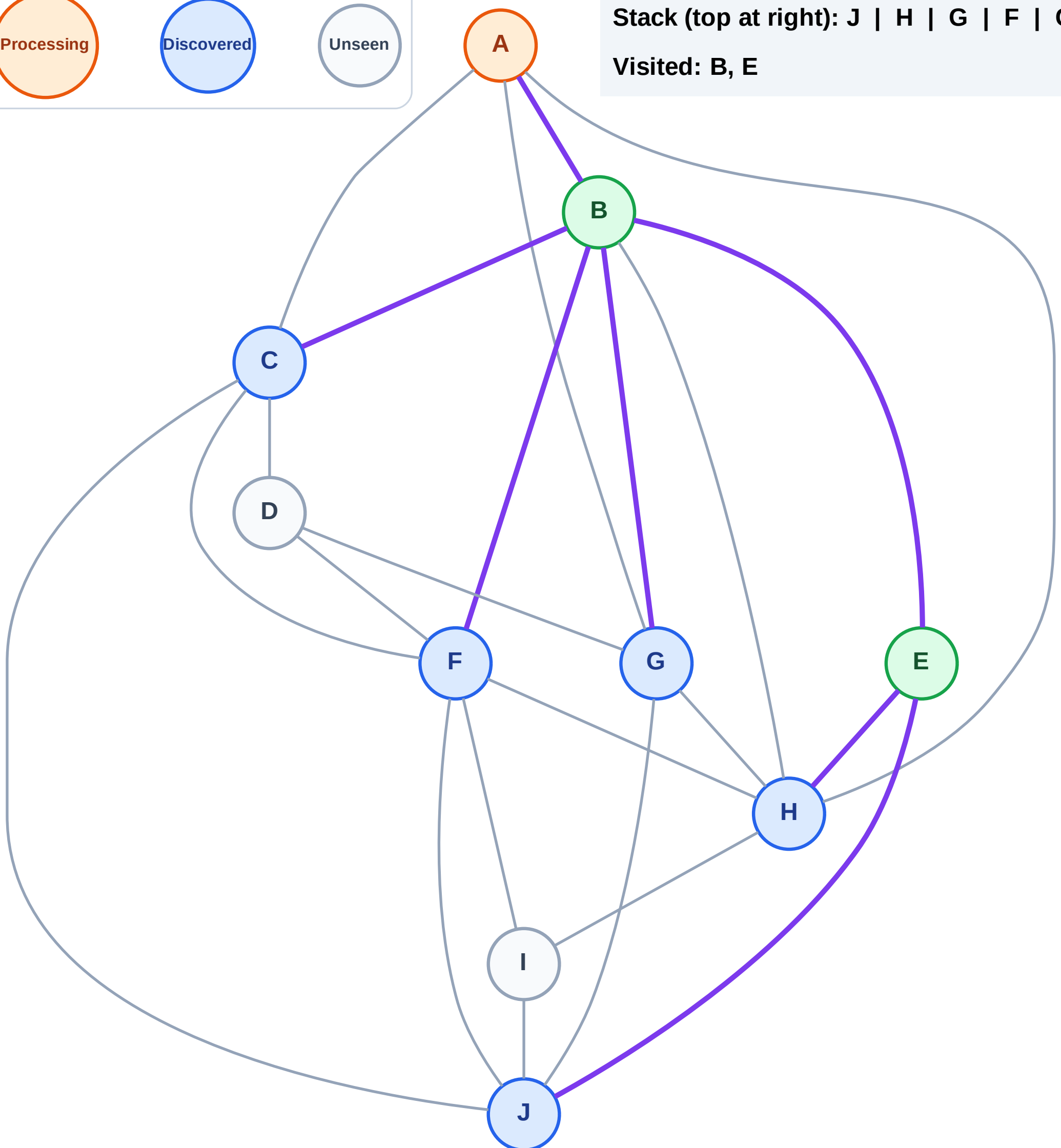
Step 6: Process node A

Legend



Stack (top at right): J | H | G | F | C

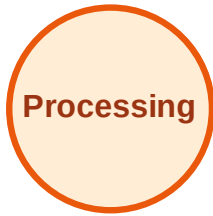
Visited: B, E



DFS Traversal

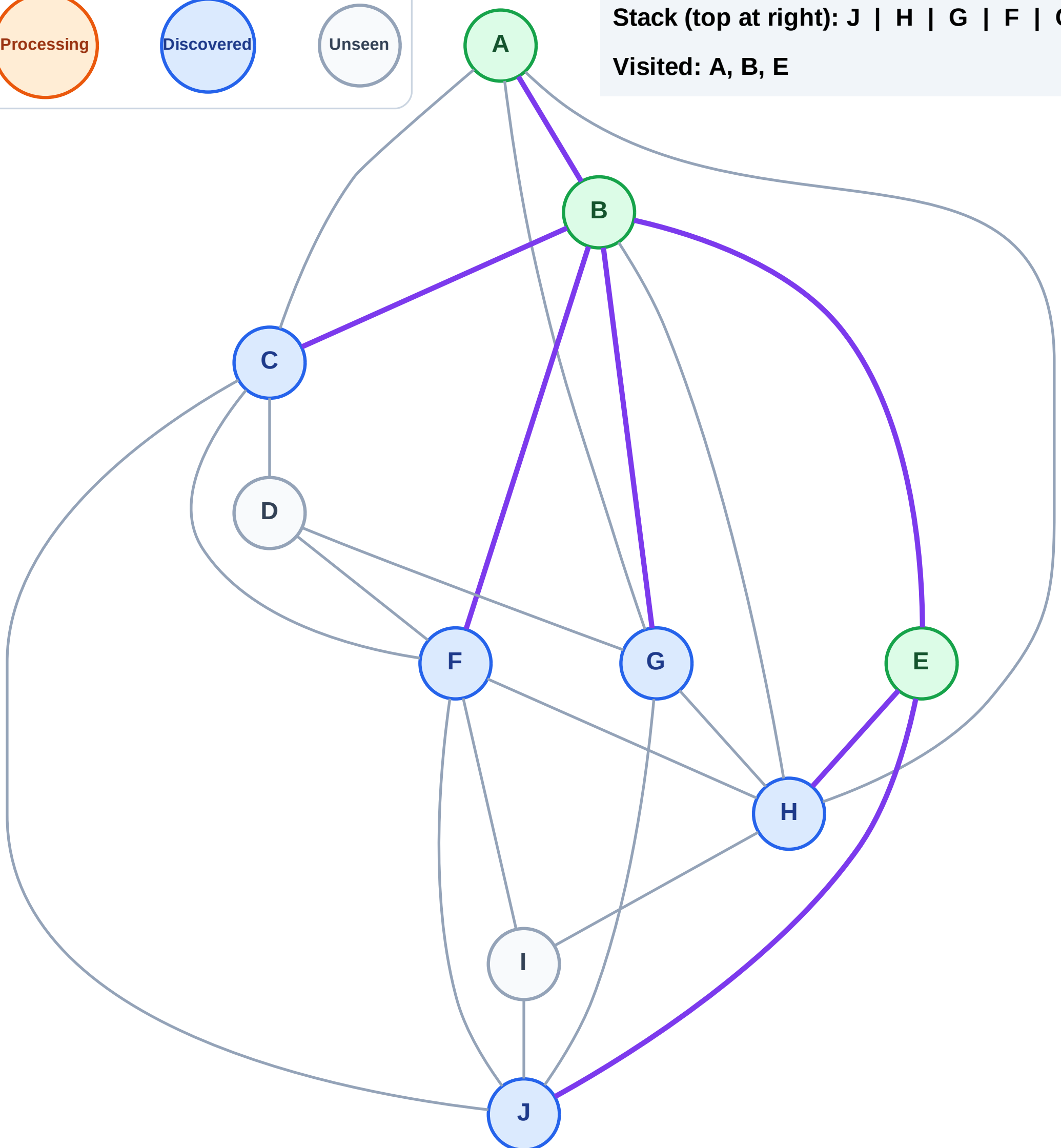
Step 7: No new neighbors from A

Legend



Stack (top at right): J | H | G | F | C

Visited: A, B, E



DFS Traversal

Step 8: Process node C

Legend

Complete

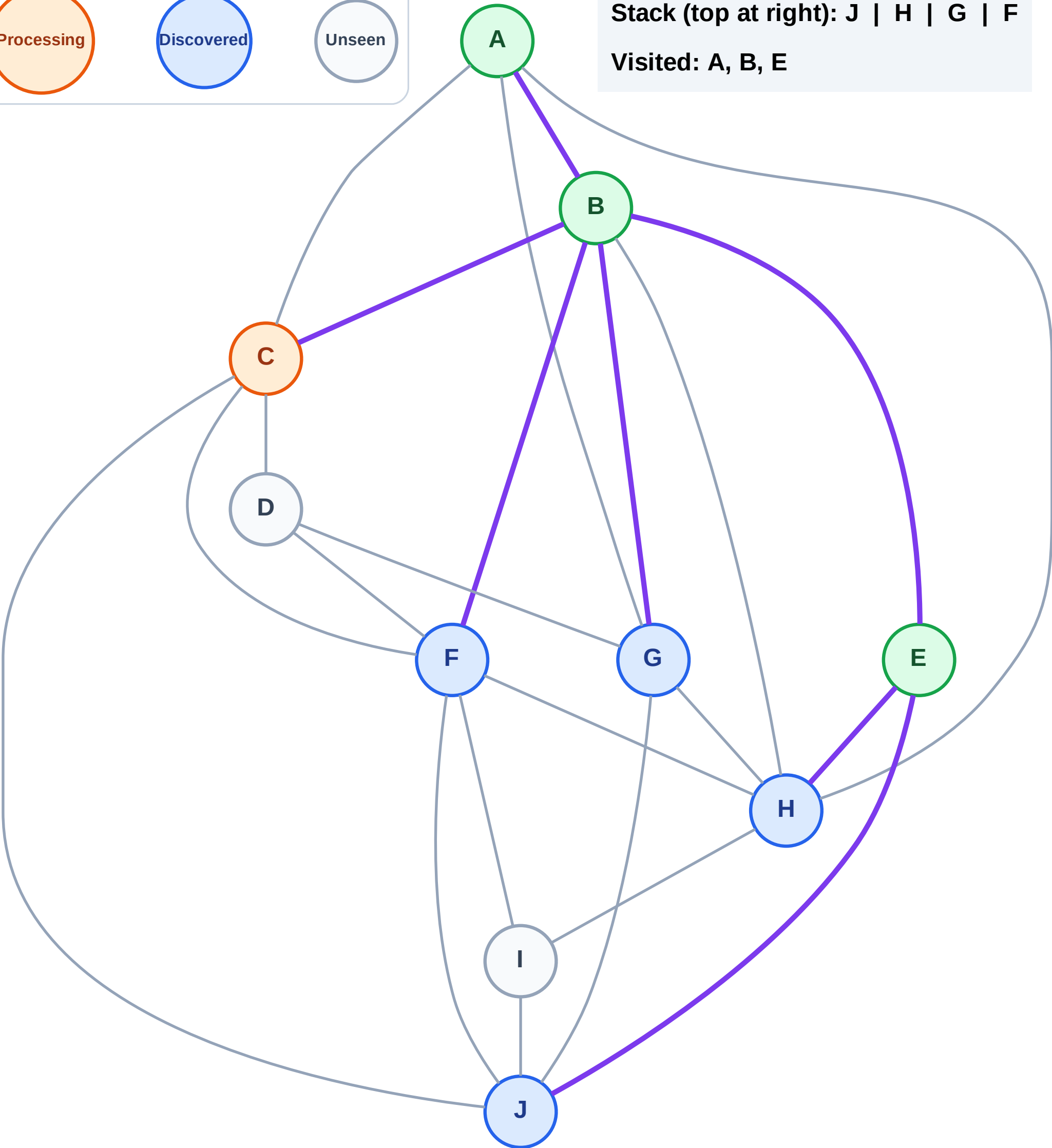
Processing

Discovered

Unseen

Stack (top at right): J | H | G | F

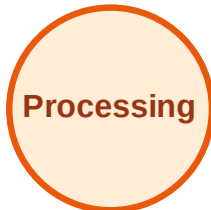
Visited: A, B, E



DFS Traversal

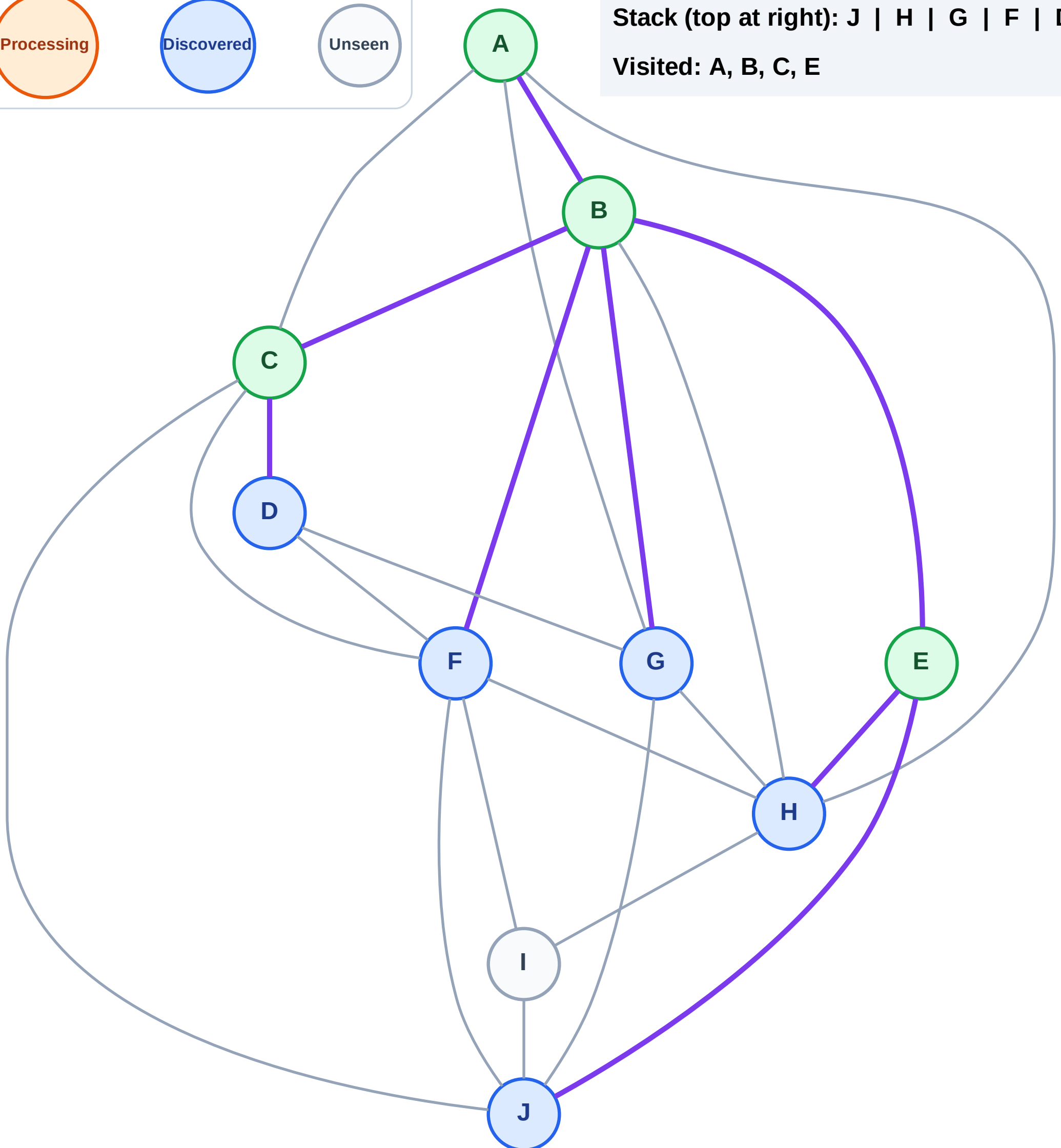
Step 9: Discover D from C

Legend



Stack (top at right): J | H | G | F | D

Visited: A, B, C, E



DFS Traversal

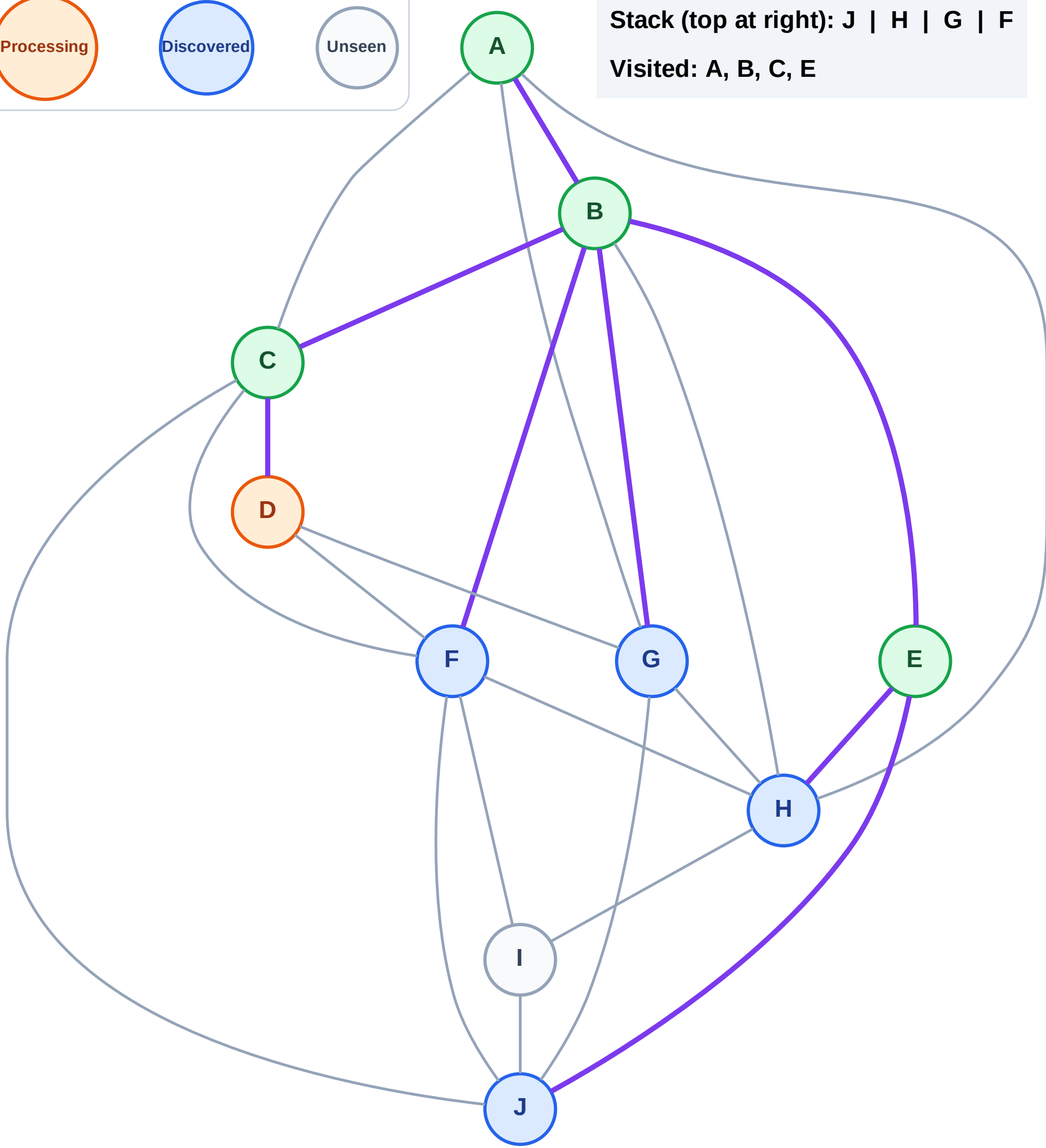
Step 10: Process node D

Legend



Stack (top at right): J | H | G | F

Visited: A, B, C, E



DFS Traversal

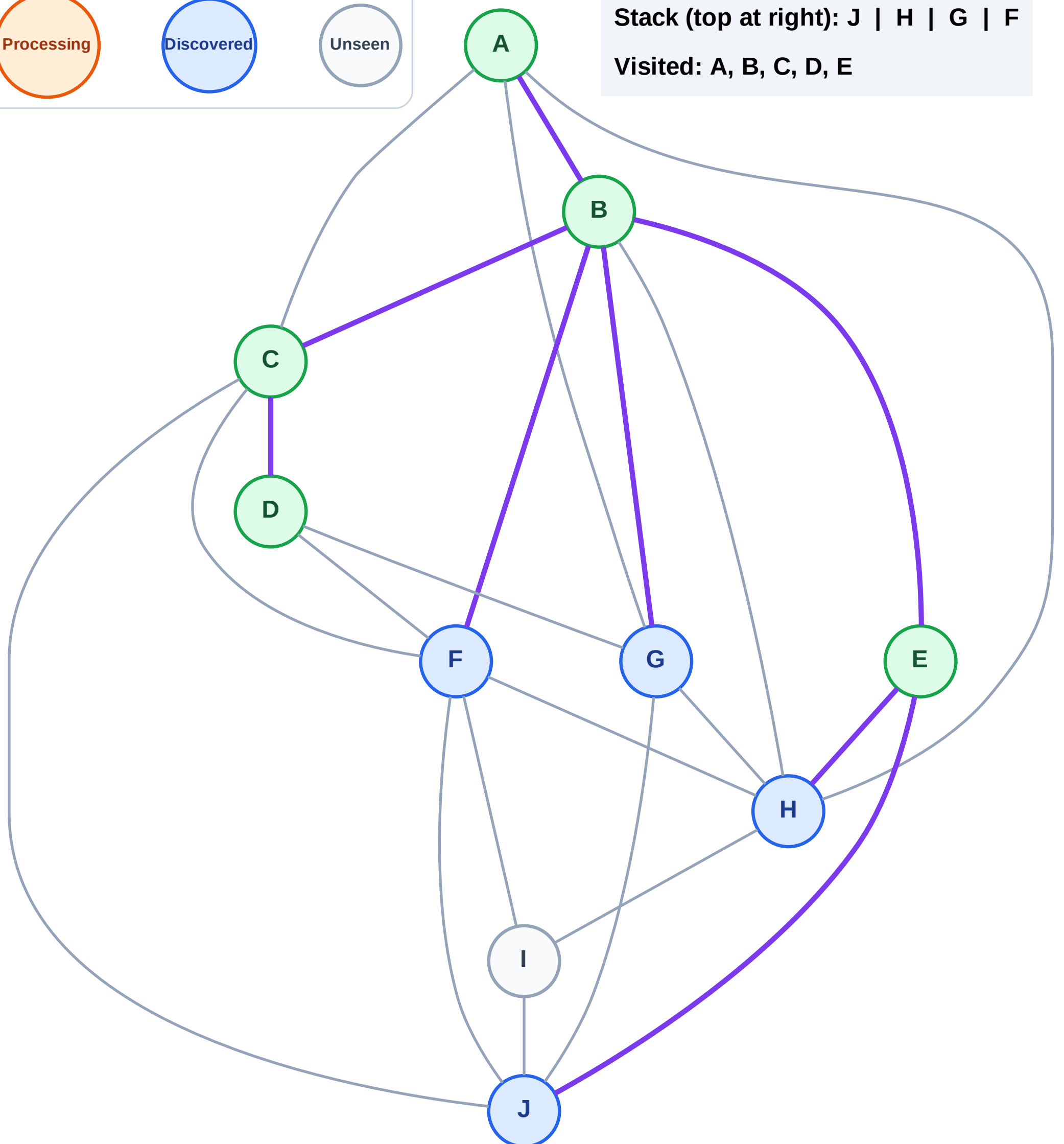
Step 11: No new neighbors from D

Legend



Stack (top at right): J | H | G | F

Visited: A, B, C, D, E



DFS Traversal

Step 12: Process node F

Legend

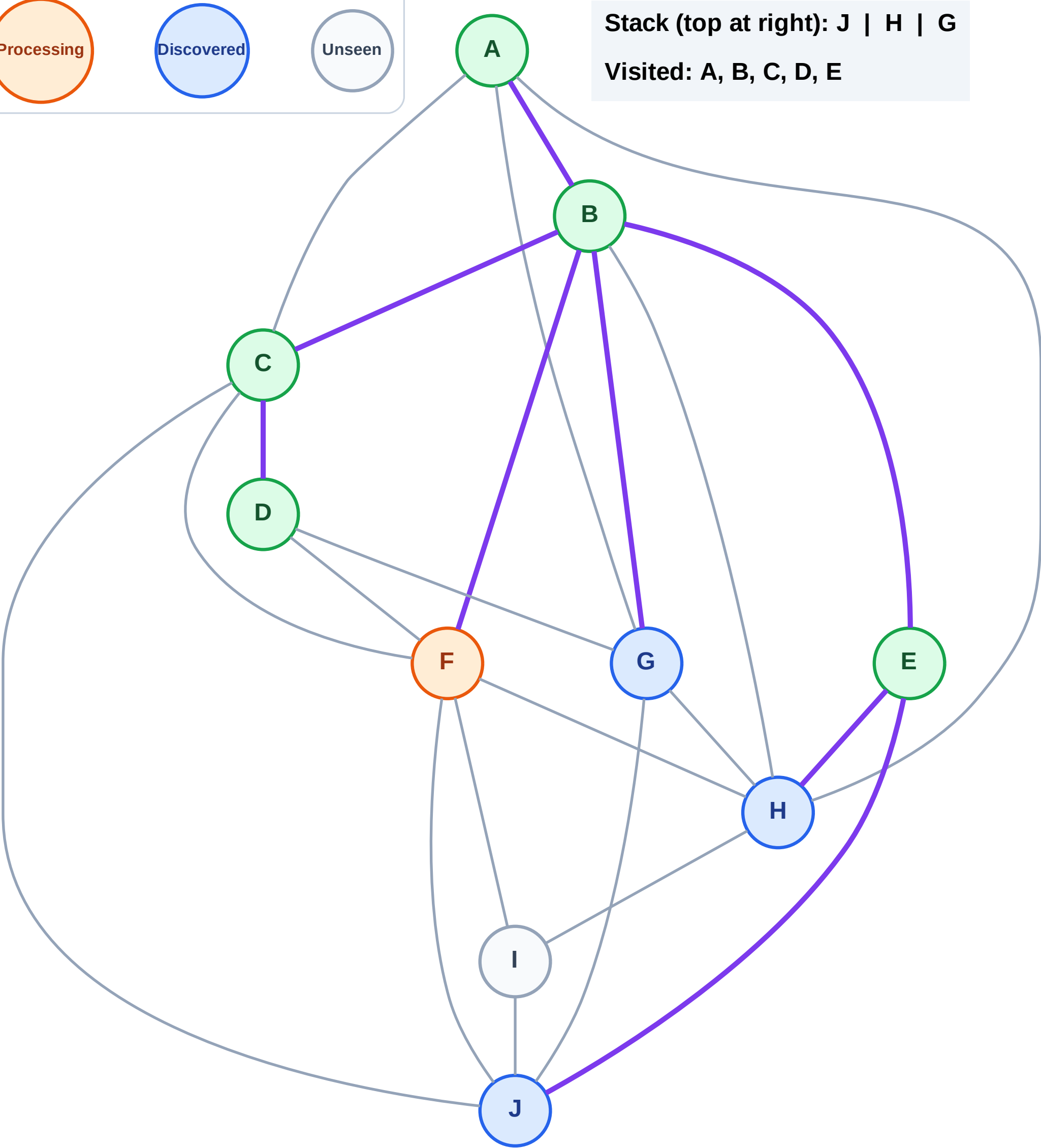
Complete

Processing

Discovered

Unseen

Stack (top at right): J | H | G
Visited: A, B, C, D, E



DFS Traversal

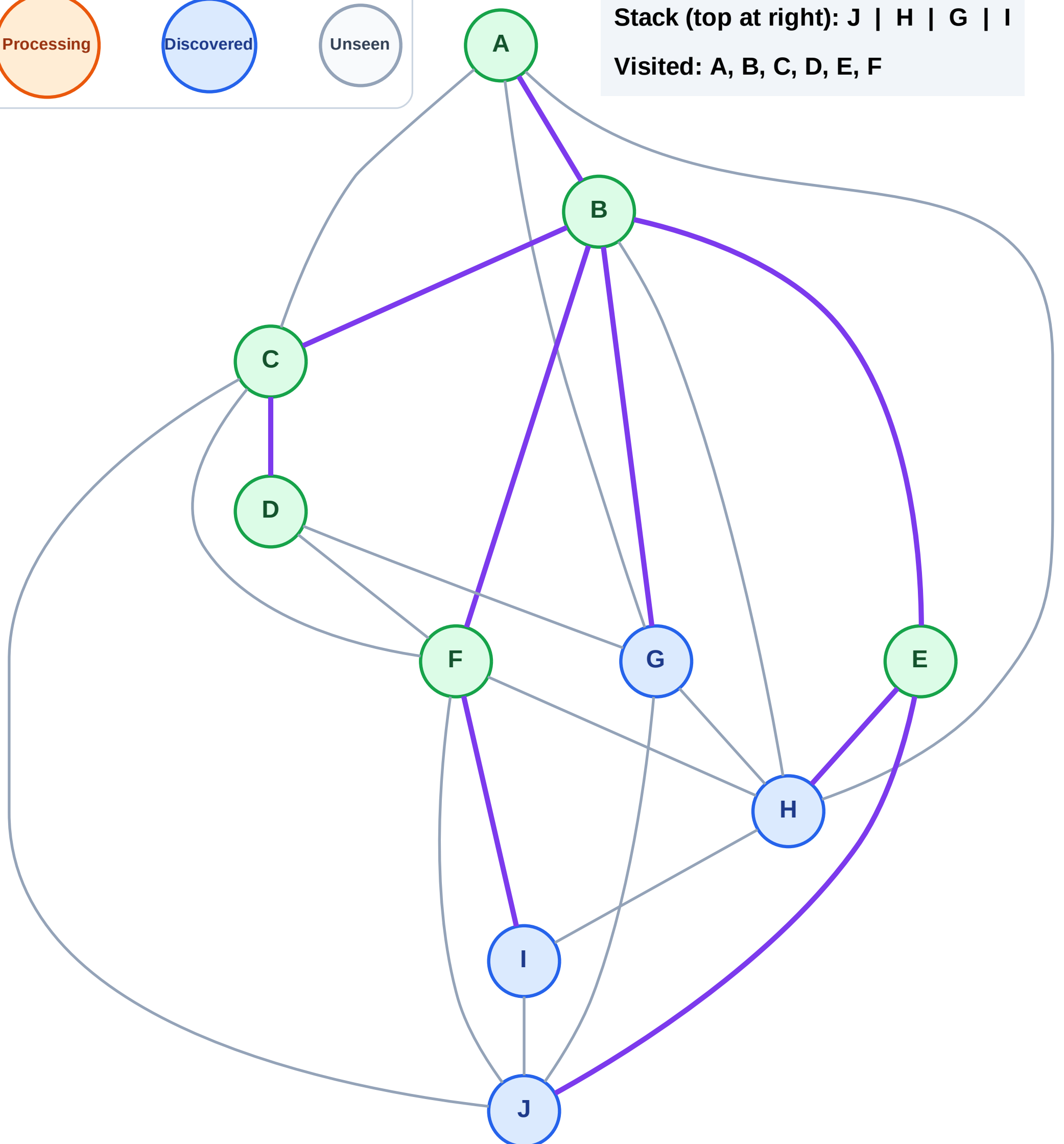
Step 13: Discover I from F

Legend



Stack (top at right): J | H | G | I

Visited: A, B, C, D, E, F



DFS Traversal

Step 14: Process node I

Legend

Complete

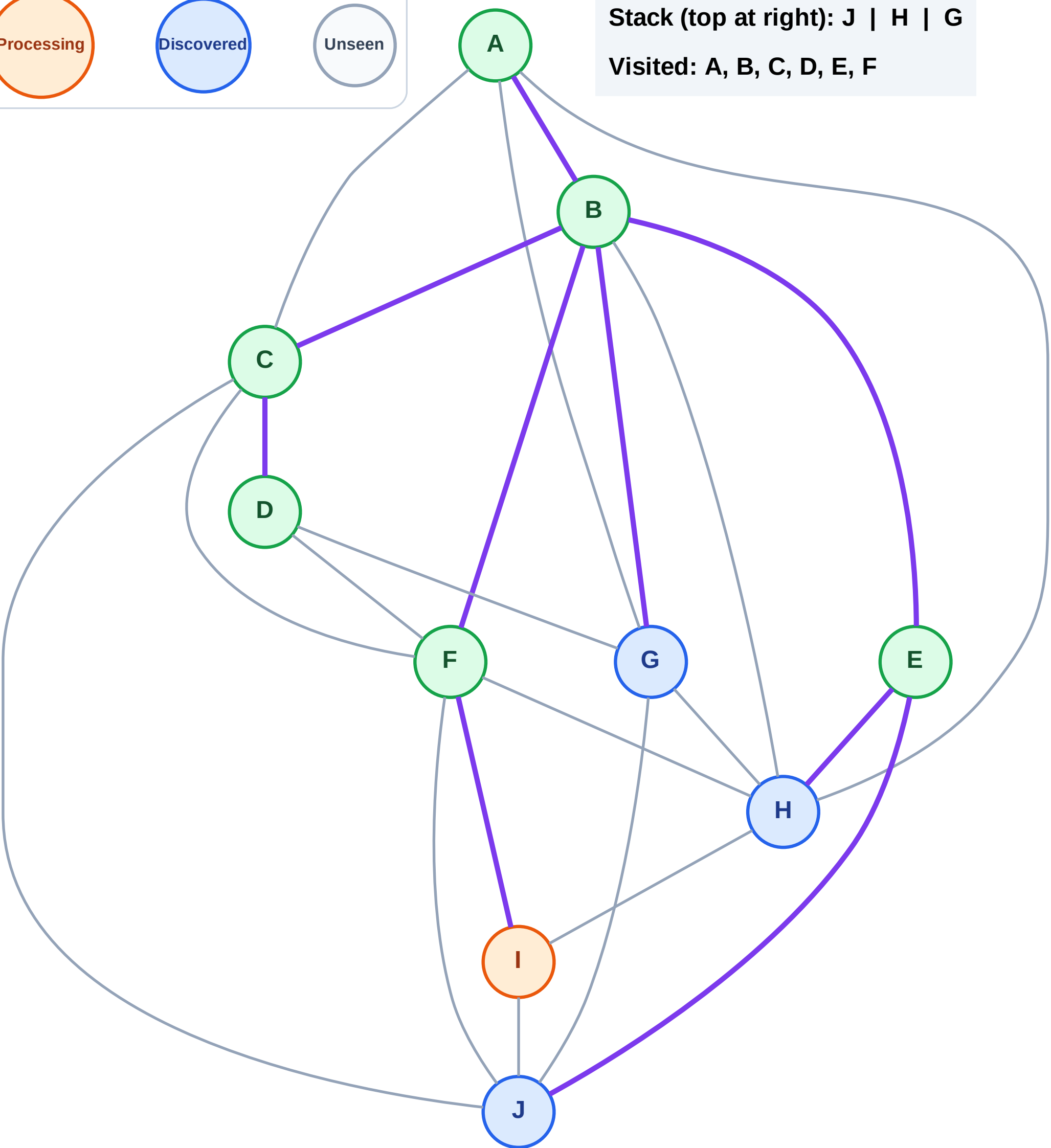
Processing

Discovered

Unseen

Stack (top at right): J | H | G

Visited: A, B, C, D, E, F



DFS Traversal

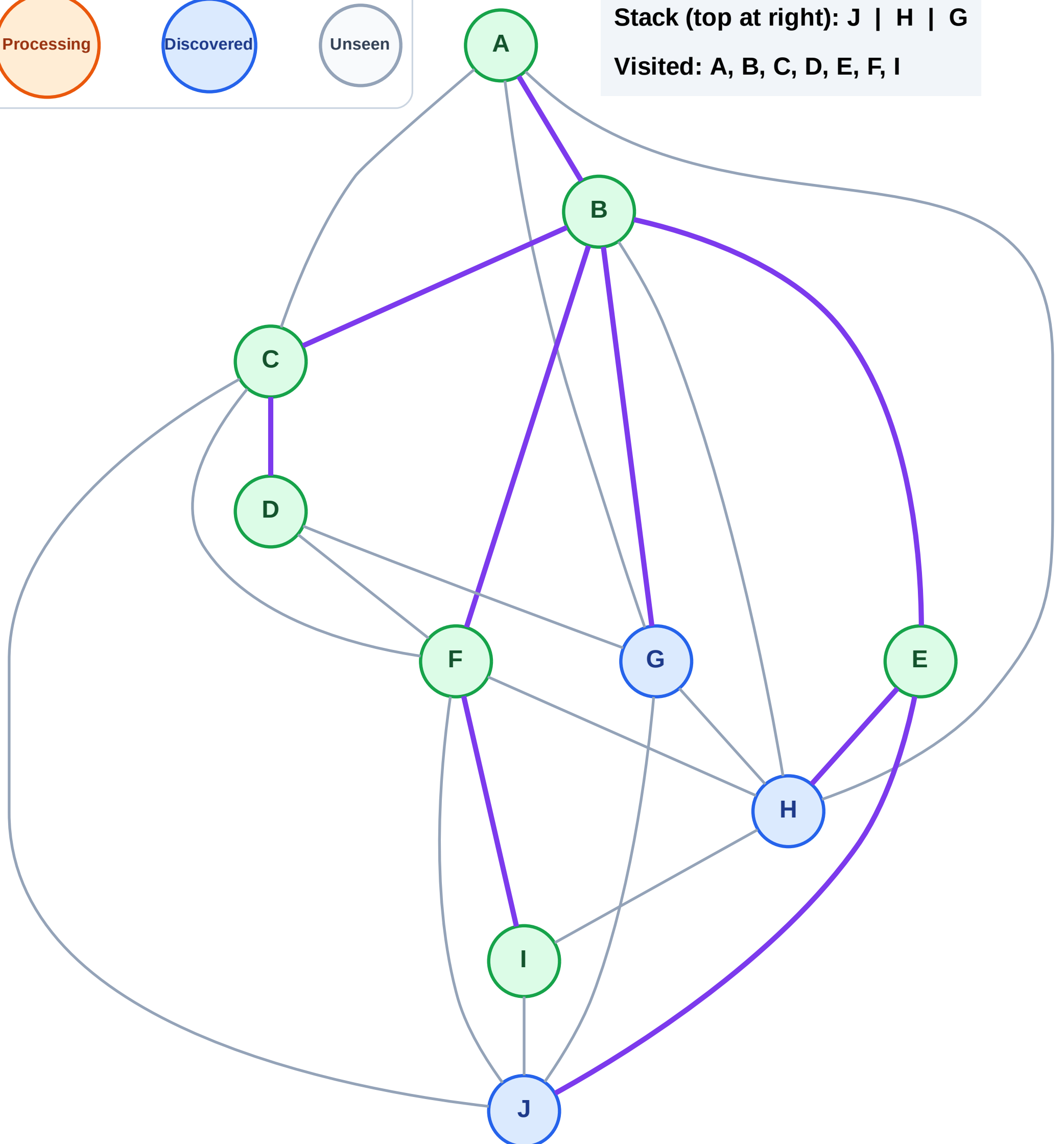
Step 15: No new neighbors from I

Legend



Stack (top at right): J | H | G

Visited: A, B, C, D, E, F, I



DFS Traversal

Step 16: Process node G

Legend

Complete

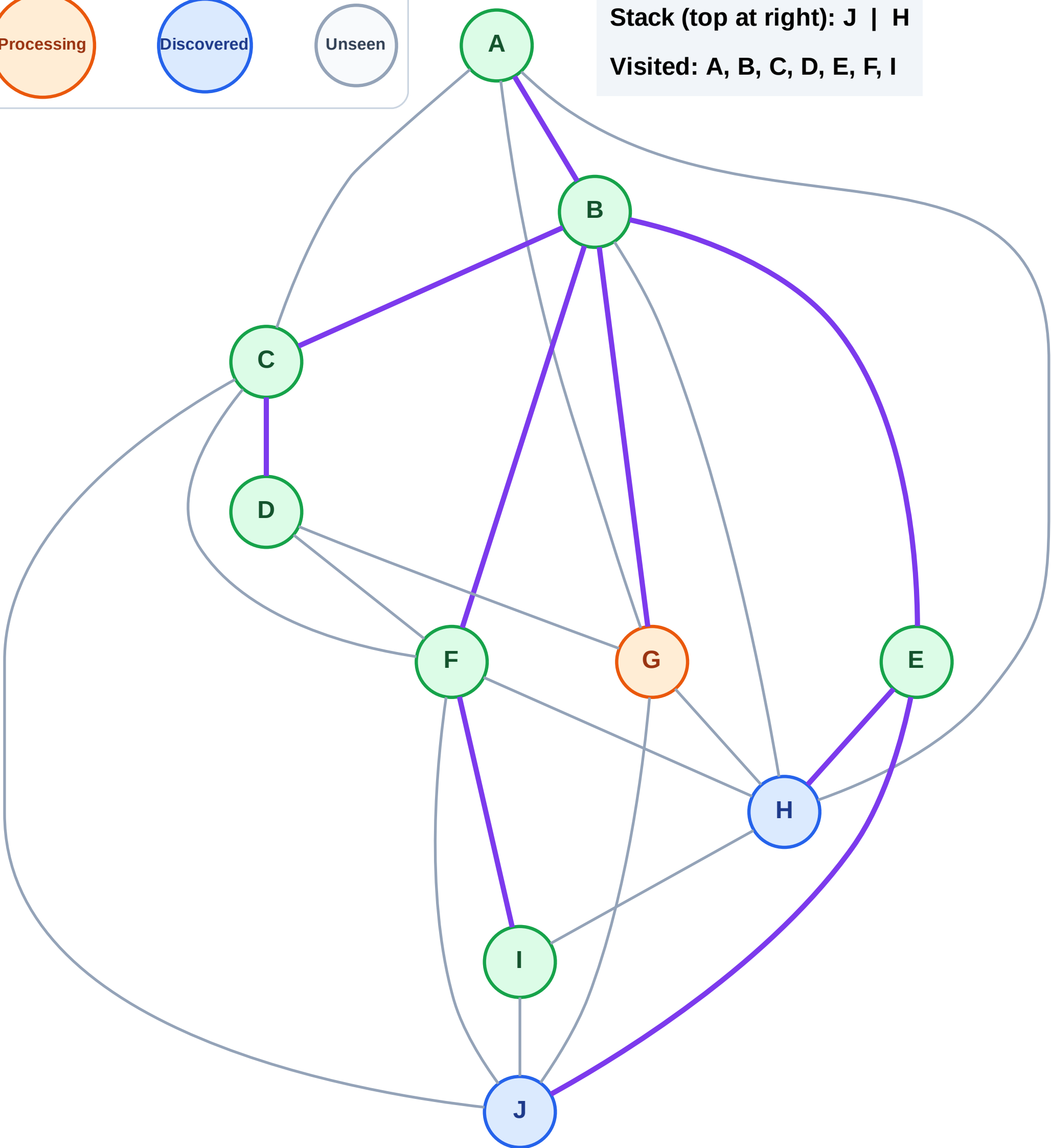
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, I



DFS Traversal

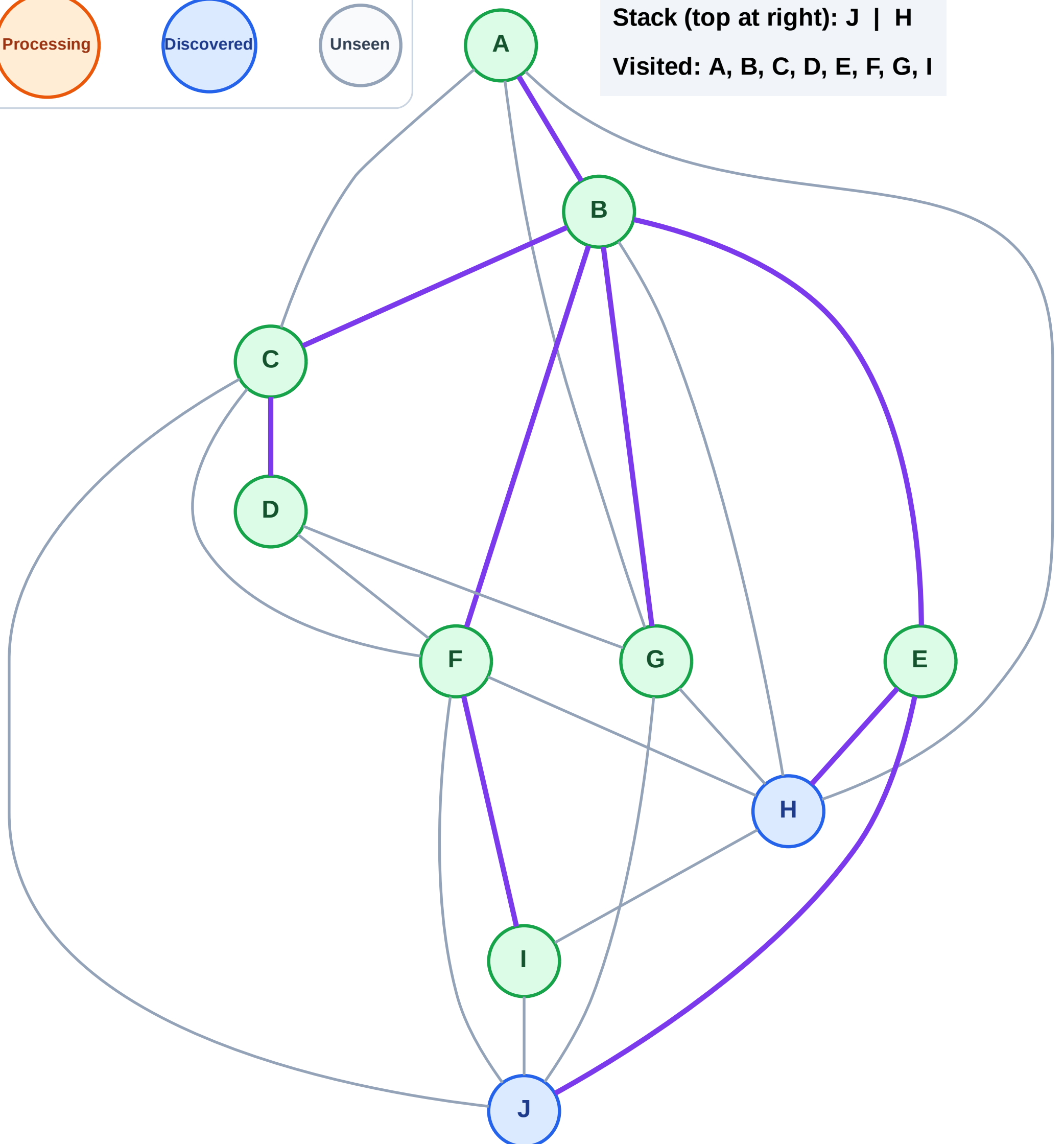
Step 17: No new neighbors from G

Legend



Stack (top at right): J | H

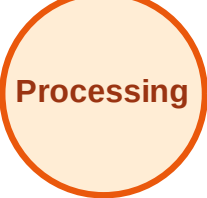
Visited: A, B, C, D, E, F, G, I



DFS Traversal

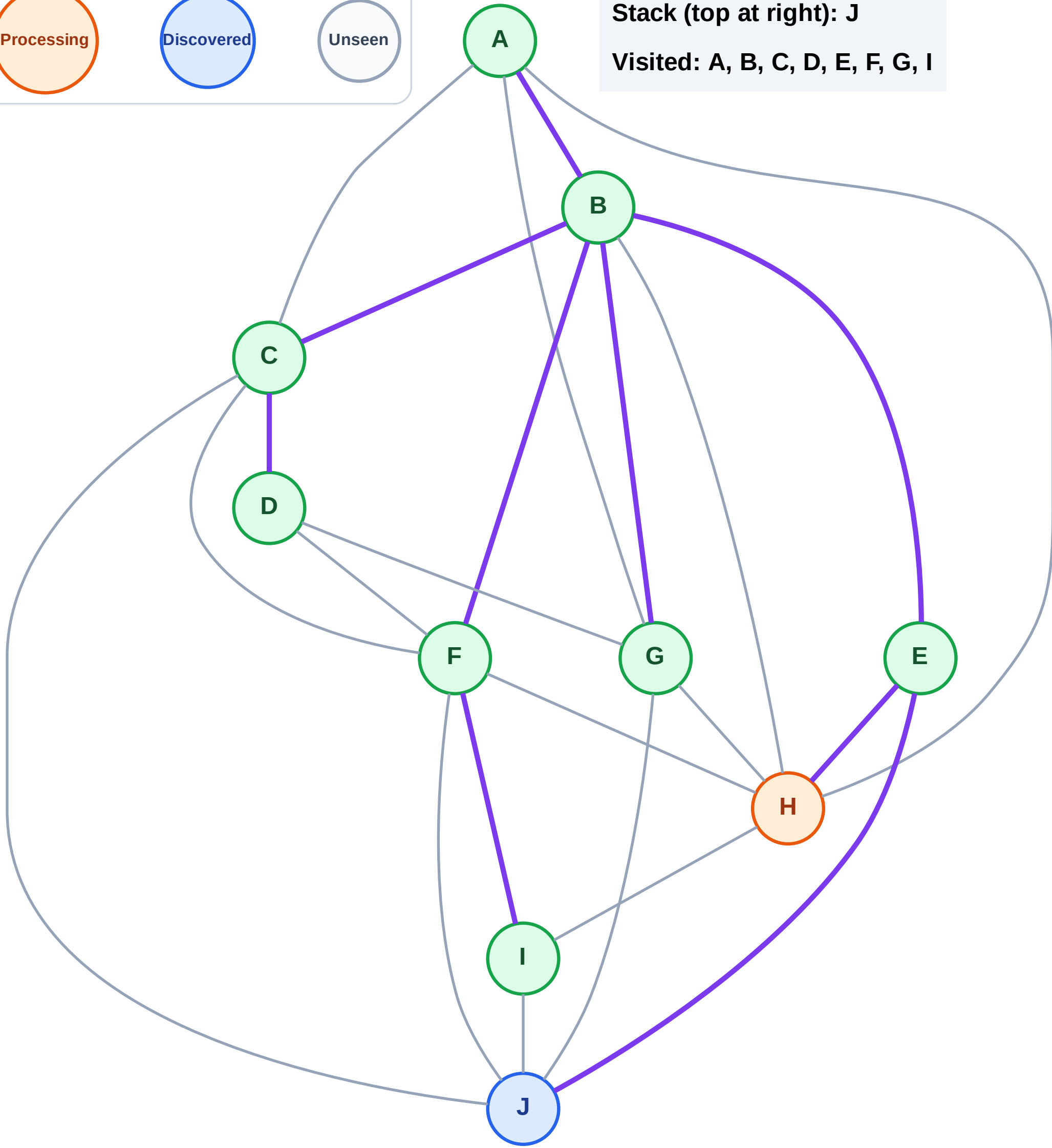
Step 18: Process node H

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, I



DFS Traversal

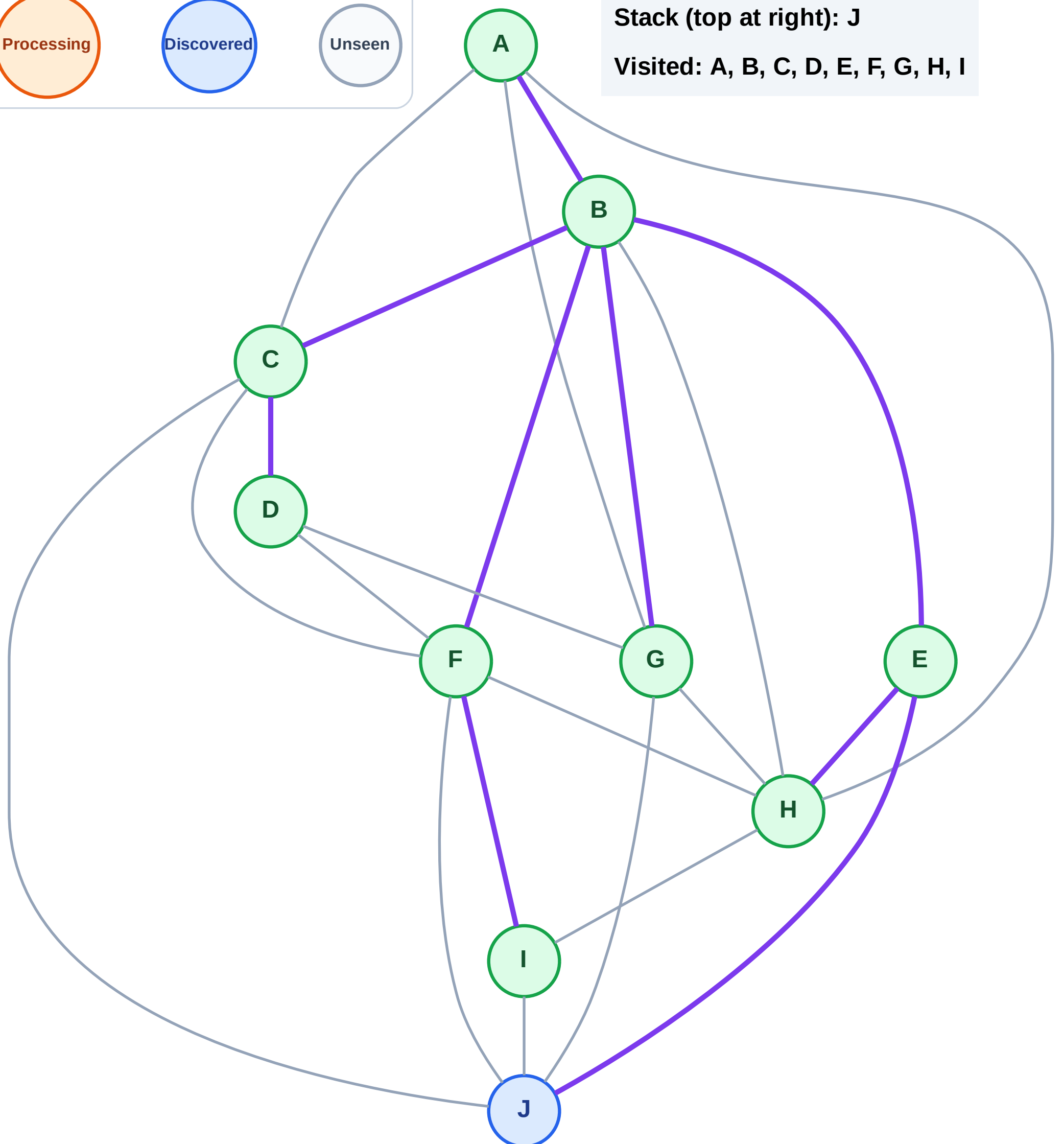
Step 19: No new neighbors from H

Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 20: Process node J

Legend

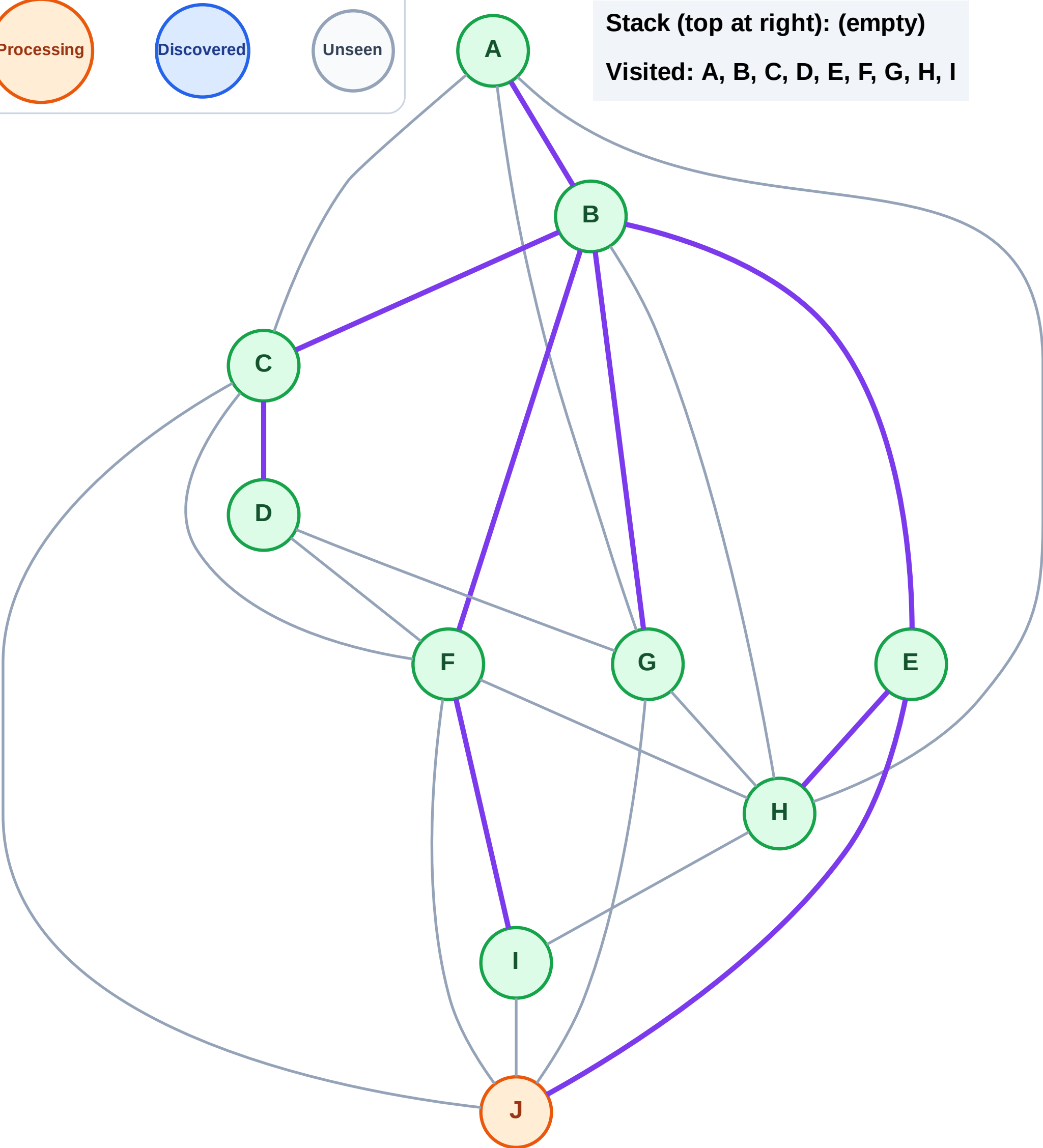
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

